



Dynamic difficulty adjustment approaches in video games: a systematic literature review

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Abstract

Providing an appropriate difficulty level in a game is critical for keeping players engaged. Dynamic Difficulty Adjustment (DDA) is a common approach for optimizing player experience by automatically modifying game aspects. This paper reviews literature addressing mechanisms for adjusting video game difficulties in response to players' performance, emotions, or personality. For this purpose, we examined DDA studies using employed machine-learning techniques, player modeling approaches, data types used to assess players' states, testbed game genre, and application. Journal and conference articles published up to September 2022 served as the data sources in this review. The findings reveal that most studies have shown significant effects of DDA on parameters such as enjoyment, flow, motivation, engagement, and immersion. In addition, machine-learning and player modeling techniques have recently received more attention in the DDA design. However, given the ever-increasing use of games in various domains, more research is needed to understand player preferences better to adjust game parameters efficiently. By conducting further research into players' cognitive characteristics, such as visual attention, working memory, and response time, it will be possible to understand players' preferences better.

Keywords Video games · Dynamic difficulty adjustment · Player modeling · Machine-learning

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1 Introduction

Game developers generally strive to maintain player engagement over a long period [1]. Video games with personalized features improve players' immersion, engagement, excitement, and challenge [2]. Several studies addressed techniques or developed games that can adapt game content, including quests [3], characters [4], and environment [5]. Additionally, some studies focused on adjusting game difficulty levels [6] based on players' features and abilities. Game adaptation can also be applied by designing an adaptive Artificial Intelligence (AI) agent that can scale or balance its difficulty level according to a human player's capability [7–9]. In other words, maintaining a high level of engagement requires having the right level of challenges suited to players' skills or affection [10]. Manipulating game difficulty can be under a player's control [11] or accomplished automatically by game engines [6]. The latter is called Dynamic Difficulty Adjustment (DDA). DDA refers to the concept of adapting games' challenges according to individual player traits, with the goal of avoiding boredom and frustration during gameplay [12].

To tackle the difficulty adjustment problem, the plan presented by Hendrix can be summarized in the following three steps [13].

- (1) In the first step, game variables related to the game difficulty should be determined. In other words, it must be defined what difficulty means in the game. Depending on the genre and purpose of the game, the difficulty is controlled differently based on various game elements [14]. For instance, difficulty can be defined based on the opponent's offensive factor or the behavior of Non-Player Characters (NPC), in a fighting game [15], the number of enemies that a player should kill in a shooter game [16], or can be considered as the number of blocks that a player should memorize in an N-Back working memory game [17]. This kind of difficulty, which includes only the game characteristics irrespective of the player, is referred to as absolute or nominal [18].
- (2) The second step is to identify variables that describe players' states, such as performance, emotion, or personality. Based on these variables, Adams proposed "perceived difficulty" as a subjective factor that depends on players' game experiences [19].

Emotion and performance can serve as indicators to estimate the level of difficulty perceived by a player. For instance, a player's score falling below a certain threshold can be indicative of the game being challenging for that individual [20]. Similarly, when a measurement tool detects a deviation in player arousal from the ideal setpoint, it can suggest that the game is perceived as challenging [21].

Moreover, player personality is also an important consideration in difficulty adjustment. A DDA system can adapt the game difficulty based on players' preferences or personality traits. By considering factors such as risk-seeking tendencies [22] or other personality characteristics [23], the DDA system can tailor the difficulty settings to align with an individual player's profile.

- (3) The last step is to determine the right difficulty adjustment method based on the variables determined in steps 1 and 2. Adjustment methods can change game difficulty based on rules or be formulated as a machine-learning problem.

Several relevant review studies have addressed the issue of adaptation and difficulty adjustment according to the three mentioned steps. For example, Dziedzic proposed three approaches for measuring game difficulty for DDA systems [24]. The author believes that there is no consistent definition for game difficulty, and according to the

genre, difficulty can be handled in different ways. Considering the second step, Paraschos systematically reviewed DDA and Procedural Content Generation (PCG) studies from 2005 to 2021 by considering assessment methods that capture aspects of players' personalities, preferences, and emotions [25]. Following a different approach, Zohaib categorized DDA approaches only concerning adjustment methods without addressing player modeling mechanisms that describe players' states [26].

Furthermore, Bontchev systematically reviewed the emotion recognition models applied to make an adaptive video game [2]. The article focuses primarily on game adaptation strategies and barely mentions DDA methods. In addition, this article does not consider other approaches that employ non-emotional data for adaptation and difficulty modification. Another example is the survey done by Lopes on the difficulties in creating an adaptive and generative gaming engine [27]. In this article, DDA methods are mentioned as one of the game adaption techniques, but it is not its main focus.

Building upon the three-step plan outlined above, we aim to provide a comprehensive analysis of the literature in the field of DDA. By applying a systematic approach, the articles encompassing DDA can be examined based on three key criteria: the game genres, data utilized to describe players' states, and methods employed for difficulty adjustment.

Additionally, it is important to consider the evaluation methods employed by the studies to assess the impact of the adjustment techniques on various dimensions of players' experiences. Evaluating the effectiveness of different DDA methods contributes to our understanding of how these techniques can enhance player engagement, enjoyment, and skill development.

Table 1, provides an overview of recent reviews on DDA, showcasing the extent to which the mentioned criteria have been thoroughly investigated. The inclusion of game genre, data utilized, adjustment method, and evaluation method columns in Table 1 allows for a comprehensive assessment of the explicitly mentioned criteria within the relevant articles. If a column was marked with an "×" it indicates that the corresponding criteria have been mentioned in the article. It becomes apparent that none of the previous articles have comprehensively examined all four criteria, which underscores the significance of our study. In comparison to the existing literature, our article places a particular emphasis on providing detailed descriptions of the machine-learning methods employed in DDA implementations.

Furthermore, in this paper, we provided game designers with a comprehensive perspective on the essential modules involved in designing and implementing DDA-based systems. By outlining the key components and their interactions, this structure facilitates a systematic approach to designing effective DDA systems that can enhance players' engagement and satisfaction.

In the landscape of most related reviews, such as the works of [25] and [26], the focus was primarily on player modeling techniques or adjustment methods. However, given the growing prominence of personalized games [29] and the increasing need for novel methods to adapt game difficulty considering individual differences, our study presents a systematic review of DDA studies that consider the application of various machine-learning techniques, player modeling approaches, and data types used for assessing players' states. Notably, apart from the work of Zohaib [26], which briefly touched upon some machine-learning techniques, previous reviews have not delved into the development of DDA methods with the aid of diverse machine-learning approaches. Additionally, in contrast to earlier studies, our research also examines DDA studies in terms of game genre and application, providing a more comprehensive analysis.

Table 1 Overview of related review articles on DDA

Study	Search Period	Game Genre	Data Utilized	Adjustment Method	Evaluation Method	Summary
[28]	2009–2021	×	×			This article specifically examined DDA methods exclusively tailored to rehabilitation games. The articles analyzed in this study relied on physiological and gameplay data. However, the evaluation methods employed by the studies to assess their DDA systems on human subjects were not reported
[25]	2005–2021	×	×		×	This article primarily centered around the assessment of player status within a DDA system through player modeling methods. However, it does not delve into the specifics of machine-learning methods. The reported games encompass a variety of genres, including serious games and entertainment games. This study does not exclusively focus on the application of DDA in different game genres and categories. The article also highlights the impact of implemented adaptation methods on players' overall experiences
[26]	2009–2018			×		This article presented a concise overview of a selected few machine-learning methods employed in DDA. However, it does not delve into player modeling methods. Additionally, there is a lack of information regarding players' data utilized and evaluation methods employed to assess their usage
[2]	2014–2015		×			The primary focus of this article is on describing adaptation methods in games, specifically emphasizing the utilization of emotional data. While it briefly touched upon DDA methods, the main emphasis is placed on the use of emotional data for adaptation purposes
[27]	Not Systematic Review (Survey)					This article briefly introduced DDA as one of the game adaptation methods but does not explore it extensively as its main focus

Thus, four main Research Questions (RQ) were followed in this study to provide insight into how the difficulty adjustment can be applied:

- (4) RQ-1: How are machine-learning and player modeling techniques involved in the DDA process?
- (5) RQ-2: What data types are employed for assessing players' states, such as performance, emotion, or personality?
- (6) RQ-3: What are the purpose and genre of the games that DDA has been implemented in?
- (7) RQ-4: How do studies evaluate their proposed DDA techniques and identify the impact of implementing adaptive difficulty on different aspects of players' experiences?

The data sources for the current systematic review were journal and conference articles published up to September 2022, as sourced via Web of Science (WOS), Scopus, IEEE Xplore, and PubMed. This review identified 85 studies that met the inclusion criteria for the final analysis. Studies were categorized based on difficulty adjustment method type, the data type used for players' states assessment, the game's purpose and genre, and the evaluation method.

The rest of the paper is structured as follows: Sect. 2 describes the systematic review methodology and selection process of relevant papers. Section 3 provides answers to the mentioned research questions and reports the results of the included papers that address this topic. Section 4 discusses a DDA system's various dimensions and challenges and indicates the potential areas and future direction. The paper is concluded in Sect. 5.

2 Method

We followed the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) statement guidelines [30]. Specifically, we adhered to the PRISMA 2020 statement, which provides a 27-item checklist and four-phase flow diagram for conducting and reporting systematic reviews and meta-analyses. Unfortunately, there is no detailed information about the quality of assessment, such as Cochrane's guidelines [31], in the included papers. Consequently, we were not able to compare them considering quality assessment measures.

2.1 Databases searched

Considering that game is a multidisciplinary field, four critical databases were queried to identify related literature: WOS, Scopus, and IEEE Xplore, which are specially dedicated to computer science and engineering, and PubMed, which is a resource for life and biological science literature.

To best address the present research questions, we selected databases that span the technical, health, and behavioral sciences. WOS and Scopus were included in our study because they are widely recognized as two of the most comprehensive and commonly used tools for literature searches [32]. Moreover, these resources encompass a broad spectrum of academic subjects [33], making them highly suitable for our research. IEEE Xplore focuses specifically on computer science and engineering literature relevant to game development. PubMed provided access to research within life and biological sciences relating to player

well-being. This combination of generalist and specialist databases ensured a reliable and multifaceted coverage of our research topic.

2.2 Search terms

Search terms were identified to cover a variety of techniques used for adapting content or adjusting levels in video games based on players' features and abilities. Accordingly, search terms should comprise three areas: terms addressing the variety of video games, terms that reflect the concept of the player model and associated challenges, and finally, terms used for adaptation and adjustment. Specifically, the following terms were searched:

((("serious" OR "interactive" OR "computer" OR "video" OR "multimedia" OR "internet" OR "digital" OR "online" OR "cognitive") AND ("game*" OR "gami*"))AND ("skill level" OR "challenge level" OR "hardness level" OR "balancing" OR "user model*" OR "player model*" OR "procedural" OR "difficulty adjust*" OR "level adjust*" OR "difficulty scaling" OR "difficulty adapt*" OR "online adapt*" OR "adaptive agent" OR "intelligent agent" OR "level generat*")) OR ("game personaliz*") OR ("dynamic game") OR ("personalized game") OR ("Self-adaptive game") OR ("adaptive gameplay") OR ("adaptive game") OR ("adjust* game hardness") OR ("adjust* game difficulty").

Search queries were conducted according to the search syntax of each database and were also limited to title and abstract data fields to produce more targeted results. The literature search included full journal and conference articles published in English before September 2022.

2.3 Inclusion and exclusion criteria

Articles had to meet the following conditions to narrow down the selected studies:

- (1) Development of Adjustment Technique: The selected articles were required to introduce a difficulty adjustment technique or explicitly mention that their proposed method could be employed for adaptation or level adjustment purposes.
- (2) Human Player Evaluation: The selected studies should have conducted an evaluation experiment involving human players to assess the effectiveness of the proposed technique.
- (3) Game Type: Our focus was on games other than exergames or physical rehabilitation games. We excluded these genres due to their unique characteristics, distinct design considerations, and specific goals related to physical activity, fitness, or therapeutic interventions.

Table 2 summarizes the inclusion and exclusion criteria used in the article selection process. Besides the above three conditions, we included journal and conference publications that developed and empirically evaluated techniques for dynamic difficulty adjustment or adaptive game level adjustment in digital entertainment, cognitive rehabilitation, and learning games. Furthermore, non-English papers and books were excluded.

Exergames and physical rehabilitation games have unique characteristics and design considerations that differentiate them from other game genres. These genres often have

Table 2 Inclusion and exclusion criteria for literature search

Criteria	Inclusion	Exclusion
Language	English	Non-English language
Year	No limitation	-
Publication Type	Journal, Conference	Theses, books, book chapters
Study Topic	Game adaptation, game level adjustment	No definition of the difficulty adjustment process
Game Type	Entertainment, cognitive rehabilitation, game-based learning	Physical rehabilitation, exergame, non-digital
Evaluation	Qualitative or quantitative experiments on human players	Simulation without testing on human players

specific goals related to physical activity, fitness, or therapeutic interventions [34]. While there may be some overlap in the underlying principles and techniques used in difficulty adjustment across different genres, the specific design considerations, goals, and player characteristics in exergames or physical rehabilitation games make their difficulty adjustment implementations unique [35]. We believed that their distinct nature warranted a separate and specialized examination, which is beyond the scope of our current study.

2.4 Screening process

After applying search terms to the mentioned databases, 12486 papers were identified. By combining references and duplicate removal, 9069 papers remained for screening. The screening is a stepwise process and is applied in three phases: title screening, abstract screening, and full-text screening [36]. After screening the titles of 9069 articles, 8044 articles were excluded. Of the 1025 remaining articles, 497 articles were excluded after the abstract screening. Example papers involving adaptation or difficulty adjustment in non-gaming domains such as manufacturing, education, user interfaces, dynamic game theory, and business management were excluded based on title and abstract screening. Thus, 528 articles were selected for eligibility assessment, from which 434 reports were discarded through inclusion criteria filtering, and 8 papers were eliminated because they could not be retrieved. Eventually, after the eligibility analysis by reading the eligible full texts several times, 85 articles met the inclusion criteria and remained for research and data extraction. PRISMA flow diagram, which describes the study selection process, is illustrated in Fig. 1.

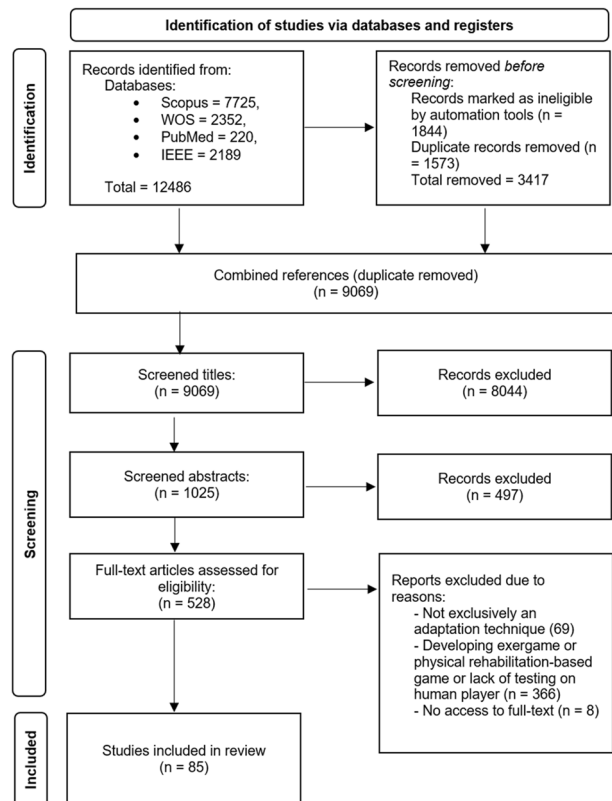
2.5 Analysis of included studies

Table 3 lists and defines extracted data from selected studies to clarify the present research questions. The approach used to apply difficulty adjustment is listed under the methodology part (RQ-1). Information about the adjustment approach provides insight into how a difficulty adjustment problem is formulated using different machine-learning approaches. Since, in several studies, player modeling approaches, closely addressed in relation to difficulty adjustment [25, 26], are also listed under the methodology part. Data details determine the type of data used in studies and identify how different data types have been acquired (RQ-2). The application item reveals the genre and purpose of games in which difficulty adjustment has been implemented (RQ-3). Finally, the evaluation section presents the participants' information and the experimental results (RQ-4). The 85 included articles were encoded using Table 3.

3 Result

In this Section, we analyzed the selected studies with respect to our research questions. Table 4 lists the details of selected studies according to items described in Sect. 2.5.

It should be noted that we extracted key literature dissemination information, including the year and publication venue, for each article. This publication information provides an overview of the types of articles primarily published in this field, specifying high-ranking journals and conferences. The publication date (year) is a valuable item that enabled us to identify trends in literature dissemination and significant developments in the game's level adjustment.

Fig. 1 PRISMA flow diagram

For brevity in the main text, publication details, a summary of the studies' objectives, the significant findings, and the conclusions of the studies are provided in Appendix 1.

3.1 Difficulty adjustment method

Before delving into our research questions, we present a categorization of the methods employed in studies to adjust game difficulty. According to the literature, game adaptation can be approached in two ways [27]. The first approach involves manipulating game parameters, such as the number of enemies or opponents' power, during gameplay. Developers design a level or difficulty adjustment system that modifies game difficulty based on players' performance, personalities, or traits [10]. The second approach personalizes the game environment or elements based on players' expectations and traits, where the selected game elements do not necessarily determine the game's difficulty [5]. Our focus is on reviewing studies that have employed the first approach. While the second approach, which involves tailoring game content to suit individual players' preferences, is an important area of research, it falls within the broader concept of game adaptation. DDA can be considered as a form of game adaptation [2], as it dynamically adjusts games' difficulties to optimize players' experiences. Additionally, the process of generating game content to player-specific information, such as gameplay preferences and traits, can be related to the field of Procedural Content Generation (PCG) [97].

Table 3 Proforma of data extraction from included articles

Category	Items		Description
Methodology	Category	Rule-based, data-driven	
	Method	Machine-learning technique and player modeling (Player Profiling (PP) or Player Experience Modeling (PEM)). This column was left blank for the rule-based category	
Data	Data type	Gameplay or psychophysiological data types	
	Psychophysiological data source	Sensors, equipment, or questionnaires used to collect psychophysiological data	
Application	Purpose	Entertainment Games (EG), Game-Based Learning (GBL), Cognitive Rehabilitation games (CRG), Other Disorders (OD)/ Single-player (SP), Multiplayer (MP)	
	Genre	Puzzle, platformer, card, board, action, adventure, simulation, fighting, etc	
Evaluation	Evaluation Instruments	Questionnaire (Q), Non-Questionnaire (N-Q)	
	Number of participants	-	

Table 4 Details of articles that conducted level adjustment

Reference	Methodology		Data		Application		Evaluation	
	Category	Method	Data Type	Psychophysiology data source	Purpose	Game Genre	Instrument	Sample Size
[37]	Rule-based		Psychophysiology	Heart rate	EG/SP	Action Adventure	Q	42
[38]	Data-driven	Case-Based Reasoning	Gameplay	-	EG/SP	Puzzle	Q	40
[39]	Data-driven	Monte Carlo Tree Search	Gameplay	-	EG/SP	Fighting	Q	20
[8]	Rule-based		Gameplay	-	EG/MP	Racing (Arcade-style)	Q	794
[40]	Data-driven	Minimax	Gameplay	-	GBL/SP	Casual	N-Q	7
[41]	Data-driven	Bayesian network	Gameplay	-	GBL/SP	Casual	Q	79
[41]	Data-driven	Bayesian network	Gameplay	-	GBL/SP	Casual	Q	79
[10]	Data-driven	Fuzzy Controller	Gameplay/ Psychophysiology	Speech	GBL/SP	Puzzle	N-Q	40
[42]	Data-driven	Reinforcement Learning	Gameplay	-	CRG/SP	Puzzle	N-Q	378
[20]	Rule-based		Gameplay	-	GBL/SP	Puzzle	N-Q	25
[43]	Data-driven	Genetic Algorithm	Gameplay	-	GBL/SP	Puzzle	N-Q	3
[4]	Data-driven	Genetic Algorithm	Gameplay	-	OD/SP	Card	Q	42
[44]	Data-driven	Reinforcement Learning	Gameplay	-	CRG/SP	Puzzle	Q	10
[45]	Data-driven	Hidden Markov Model	Gameplay	-	EG/SP	Shooting	Q, N-Q	11
[46]*	Data-driven	PEM / DNN	Gameplay	-	EG/MP	Fighting	N-Q	213
[47]	Data-driven	Reinforcement Learning	Gameplay	-	EG/SP	Fighting	Q, N-Q	4,10
[48]	Data-driven	LSTM-DNN	Gameplay	-	EG/SP	Puzzle	N-Q	56
[5]*	Data-driven	PP / Matrix Factorization	Psychophysiology	EDA	GBL/SP	Puzzle	Q, N-Q	266
[49]*	Data-driven	PP / Questionnaire-Based	Psychophysiology	Questionnaire	GBL/SP	Platformer	Q, N-Q	51
[50]	Data-driven	Adaptive Neuro-Fuzzy Inference	Gameplay	-	CRG/SP	Puzzle	Q, N-Q	12
[51]	Data-driven	Fuzzy Controller	Gameplay	-	GBL/SP	Puzzle Adventure	Q	60

Table 4 (continued)

Reference	Methodology		Data		Application		Evaluation	
	Category	Method	Data Type	Psychophysiology data source	Purpose	Game Genre	Instrument	Sample Size
[13]	Rule-based		Gameplay	-	GBL/SP	Puzzle, Fighting	Q	45
[6]*	Data-driven	PEM / Random Forest	Psychophysiology	Facial Expression	EG/SP	Platformer	Q	25
[52]	Rule-based		Gameplay	-	GBL/SP	Puzzle	N-Q	101
[53]	Rule-based		Gameplay	-	EG/SP	Action Adventure	Q	30
[54]	Rule-based		Gameplay	-	EG/SP	Fighting	Q, N-Q	19
[55]	Rule-based		Gameplay	-	EG/SP	Shooting	N-Q	5
[56]	Data-driven	Monte Carlo Tree Search	Gameplay	-	EG/SP	Fighting	Q, N-Q	31
[57]	Rule-based		Gameplay	-	EG/SP	Puzzle	N-Q	138
[58]	Data-driven	Case-Based Reasoning	Gameplay	-	EG/SP	Puzzle	Q	26
[59]	Data-driven	Logit Regression	Gameplay	-	EG/SP	Shooting	N-Q	37
[60]*	Data-driven	PP / Questionnaire-Based	Psychophysiology	Questionnaire	GBL/SP	Puzzle	Q	64
[61]	Rule-based		Gameplay	-	GBL/SP	Puzzle	Q	10
[62]	Rule-based		Psychophysiology	Questionnaire	EG/SP	Platformer	Q	66
[63]	Rule-based		Psychophysiology	Facial Expression	EG	Action Adventure	Q	32
[64]	Data-driven	Monte Carlo Tree Search	Gameplay	-	EG	Fighting	Q	38
[65]	Rule-based		Gameplay/ Psychophysiology	EEG	EG/MP	Shooting	Q	24
[66]*	Data-driven	PEM / Trace Theory	Gameplay/ Psychophysiology	Eye Gaze	EG/MP	Racing (Driving simulator, Like an AAA-style game)	Q	52
[67]	Rule-based		Gameplay/ Psychophysiology	EEG	EG/SP	Platformer	N-Q	25
[68]	Rule-based		Gameplay	-	EG/SP	Fighting	Q	30
[69]	Rule-based		Gameplay	-	EG/SP	Puzzle	Q	30

Table 4 (continued)

Reference	Methodology		Data		Application		Evaluation	
	Category	Method	Data Type	Psychophysiology data source	Purpose	Game Genre	Instrument	Sample Size
[70]	Data-driven	Probabilistic Graph	Gameplay	-	EG/SP	Board	N-Q	140,000
[71]	Data-driven	Monte Carlo Tree Search	Gameplay	-	EG/SP	Fighting	Q	2
[72]	Rule-based		Gameplay	-	EG/MP	Real-Time Strategy	Q	11
[73]	Rule-based		Psychophysiology	EEG	EG/SP	Puzzle	Q	10
[74]	Data-driven	Reinforcement Learning	Gameplay	-	EG/MP	Racing	N-Q	30
[14]	Data-driven	Regression Model	Psychophysiology	Questionnaire	EG/SP	Shooting	N-Q	160
[75]	Rule-based		Psychophysiology	EEG	GBL/SP	Simulation	Q	8
[16]	Data-driven	Genetic Algorithm	Gameplay	-	EG/SP	Platformer	N-Q	17
[76]	Rule-based		Gameplay	-	EG/SP	Casual	Q	40
[21]	Data-driven	Genetic Algorithm	Gameplay	-	EG/SP	Puzzle	N-Q	
[77]	Data-driven	Control Theory Laws	Psychophysiology	EDA	EG/SP	Racing (Arcade-style)	N-Q	25
[78]	Rule-based	N-Gram	Gameplay	-	EG/SP	Board	Q	37
[79]	Rule-based		Psychophysiology	Attention	EG/SP	Adventure	N-Q	40
[17]	Rule-based		Gameplay	-	CRG/SP	Puzzle	Q, N-Q	21
[80]*	Data-driven	PEM / Tensor Factorization	Gameplay	-	CRG/SP	Puzzle	Q, N-Q	14
[81]	Rule-based		Gameplay	-	EG/SP	Role-Playing	N-Q	63
[82]*	Data-driven	PEM / Bayesian Network	Gameplay	-	EG/SP	Shooting	Q	42
[83]	Rule-based		Psychophysiology	Questionnaire	GBL/SP	Simulation	Q, N-Q	10
[84]	Rule-based		Gameplay	-	EG/SP	Platformer	Q	58
[85]	Rule-based		Gameplay	-	EG/SP	Platformer	Q	30
[86]	Rule-based		Psychophysiology	EDA, GSR	OD/SP	Shooting	N-Q	9
					EG/MP	Shooting	Q, N-Q	62

Table 4 (continued)

Reference	Methodology		Data		Application		Evaluation	
	Category	Method	Data Type	Psychophysiology data source	Purpose	Game Genre	Instrument	Sample Size
[87]	Rule-based		Gameplay	-	EG/MP	Racing (Arcade-style)	Q, N-Q	30
[88]	Rule-based		Gameplay	-	CRG/SP	Adventure	Q	24
[89]	Rule-based		Gameplay	-	EG/SP	Casual	Q	90
[90]*	Data-driven	PEM / Kalman Filter	Psychophysiology	Facial Expression	EG/SP	Puzzle	N-Q	20
[91]*	Data-driven	PEM / Neuroevolutionary Preference Learning	Gameplay/ Psychophysiology	Head Movement	EG/SP	Platformer	N-Q	58
[22]*	Data-driven	PP / Particle Filter	Gameplay		EG/SP	Puzzle	N-Q	59
[92]*	Data-driven	PEM / Singular Value Decomposition	Gameplay	-	EG/SP	Shooting	N-Q	210
[3]*	Data-driven	PEM / Tensor Factorization	Gameplay	-	EG/SP	Role-Playing	N-Q	32
[93]*	Data-driven	PP / Questionnaire-Based	Psychophysiology	Questionnaire	EG/SP	Shooting	Q	21
[94]	Rule-based		Gameplay/ Psychophysiology	Questionnaire	EG/SP	Shooting	Q	57
[95]	Rule-based		Psychophysiology	ECG, Skin Conductance, respiration	EG/SP	Puzzle	Q, N-Q	15
[15]	Data-driven	Minimax	Gameplay	-	EG/SP	Board	Q	50
[96]	Rule-based		Psychophysiology	Facial Thermal Signature	EG/SP	Shooting	Q	14

Difficulty adjustment and PCG are related but they are distinct concepts in game design and development. The former centers around real-time adaptation of game challenges to ensure an optimal and engaging experience for players. On the other hand, PCG involves algorithmically generating game content, including levels, maps, quests, and items, tailored to individual players' preferences, abilities, or characteristics.

The difficulty adjustment method indicates how should the difficulty of the game change. In general, by analyzing the related studies, the following categorization, which is presented in Fig. 2, can be summarized:

In the offline category, the difficulty of the game is fixed and predefined and provided to the player through traditional menu-based representation (e.g., easy, medium, hard) [24]. The difficulty level is changed during gameplay in the real-time category. In the manual category, known as player-oriented, which we will refer to as player-adjusted [11], players are allowed to change the game's difficulty through the game mechanics [98]. There are two strategies used in automatic methods: Static and dynamic. In the former, game designers construct a static difficulty curve that defines how a game's difficulty changes based on time or distance traveled throughout the gameplay regardless of players' differences [14]. In the latter, the adjustment method suggests that a game should be individualized for a particular player and provides automatic and real-time difficulty adjusting based on players' abilities or affective states, known as Dynamic Difficulty Adjustment (DDA) [48]. In this study, we divided DDA approaches into rule-based and data-driven methods.

The categorization proposed in this article exhibits noteworthy similarities to Soderman's framework for difficulty adjustment [99]. In the following, we explain Soderman's category and its relation to our categorization:

- (1) Design-centric flow: involves the utilization of static challenges that increase in preprogrammed or linear modes. In the preprogrammed mode, players have a small degree of control to change the difficulty of a game. For example, they may select easy, medium, or hard modes. In the linear mode, the challenge of a game increases as the game progresses in a predetermined schedule. In our categorization, the preprogrammed mode is equivalent to our "offline" approach. It aligns with the idea that game designers provide difficulty to players through traditional menu-based representation. On the other hand,

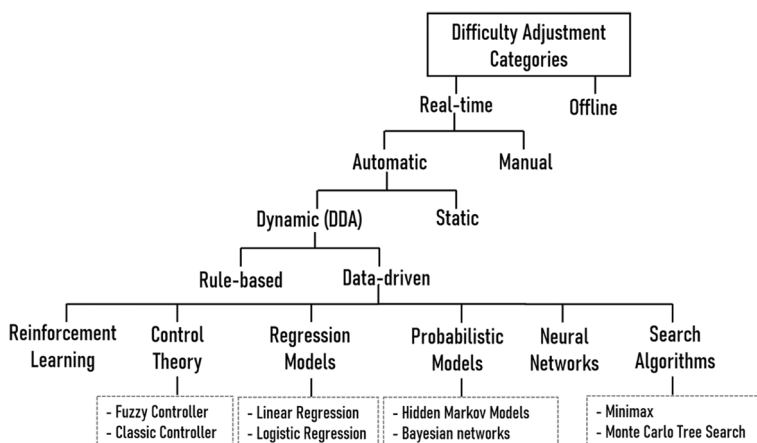


Fig. 2 Categorization of difficulty adjustment approaches

in our categorization, the equivalent to the linear mode is referred to as the “static” mode. It aligns with the idea that game designers strategically design a static difficulty curve to provide a smooth and gradually increasing challenge for players.

- (2) Player-centric flow: Soderman highlights the importance of player choices in balancing skills and challenges within the game [99]. Unlike traditional menu-based methods where difficulty adjustments are made via menus before the game commences, this approach allows players to alter game elements during gameplay itself. In our classification, we refer to this approach as a “manual” or “player-adjusted” approach.
- (3) Adaptive flow: is equivalent to our “dynamic” approach. It emphasizes the use of data mining and algorithms to dynamically adjust games’ difficulties based on players’ skill levels, enabling real-time adaptation and personalized challenges.

The majority of included papers focused on implementing a DDA system and assumed player-adjusted or static methods as a baseline for evaluation. Hence, the main focus of the present review is to investigate the research on DDA techniques which are explained in the sections below.

3.1.1 DDA methods

A growing number of studies have developed DDA mechanisms to adapt game challenges tailored to individual traits. Many of these methods take advantage of the flow theory developed by Csikszentmihalyi [12], and in the context of a serious game, they leverage the Zone of Proximal Development (ZPD) theory [100]. Flow theory postulates that matching the task difficulty to players’ abilities can improve motivation [12]. The player is neither frustrated if the game is too challenging nor bored if it is too easy. Furthermore, instead of considering players’ abilities, some methods attempt to detect players’ affective states and provide a proper game challenge accordingly. Section 3.3 discusses all the criteria used in the adjustment process to describe various aspects of players’ states.

DDA techniques can change games using different methods divided into two general classes, i.e., rule-based or data-driven approaches, to specify how parameters should change according to players’ states.

Rule-based DDA Rule-based systems are accordant with designers’ intuitions, which define simple rules regarding players’ scores to change the game difficulty. Then during the gameplay, the game’s difficulty is changed if a particular rule is fired. In other words, the one joint facet in DDA methods is to investigate a measure that indicates the player’s level of challenge. For this purpose, the player’s performance tracks in the rule-based DDA in any given instant. A decision on applying DDA can be taken by comparing the player score or response time to a threshold value or calculated using a challenge function.

For instance, as listed in Table 4, Jácome-Amores developed a gamification tool to improve children’s mathematics learning in which changing game difficulty was based on the number of achieved stars by players [20]. This rule-based DDA reduces or increases the level of difficulty between two consecutive levels based on the child’s performance.

In the context of entertainment games, Denisova changed the game difficulty for a shooting game by modifying the game timer [81]. They evaluated the effects of playing the game with an adjusted timer compared to the standard timer on player immersion. The results show higher immersion when playing the game with an altered timer compared to the standard timer.

Several studies have been applying rule-based DDA mechanisms in cognitive rehabilitation games. Nagle described a study that increased or decreased N in N-back memory tasks based on players' errors [79]. A rule-based DDA-visual adjustment method has been proposed in which visual elements such as background, foreground, and animation are changed based on players' success or failure rates [17].

Several studies introduced heuristic challenge functions based on player performances to swap between different difficulty levels (e.g. [54, 71]). Furthermore, numerous studies investigated the influence of using rule-based DDA approaches on player experience [69], enjoyment [88], skill [52], and confidence [57] compared to a standard non-adaptive version of the game.

Furthermore, several studies also use psychophysiological data to provide better insight into the player game experience. Kouliris presented a high-level saliency map to predict player attention and embedded this predictor in the game for changing difficulty level with the help of repositioning, tinting, or changing game elements' visibility [78]. Ewing proposed a study that extracted two bands of EEG signal, i.e., frontal theta and parietal alpha, and used them to measure motivation and effort [72]. Then, players' effort and motivation are compared to a baseline value, and their game's difficulty is adjusted based on increasing or decreasing this difference. Further usage of psychophysiological measures has been mentioned in Sect. 3.3.

Data-driven DDA Data-driven approaches employed statistical and machine-learning techniques to adjust game difficulties and model players' experience. Machine-learning, a subset of AI, focuses on enabling computers or systems to learn and improve from data without explicit programming. In these approaches, a model of players' psychophysiology, such as players' emotions or personality traits, is also created to predict several aspects of player experience. These techniques are explained in Sect. 3.2.

3.2 RQ-1: How are machine-learning and player modeling techniques involved in the DDA process?

Machine-learning and player modeling techniques have been categorized as data-driven approaches to adjusting game difficulty and model player experience. Figure 3 illustrates

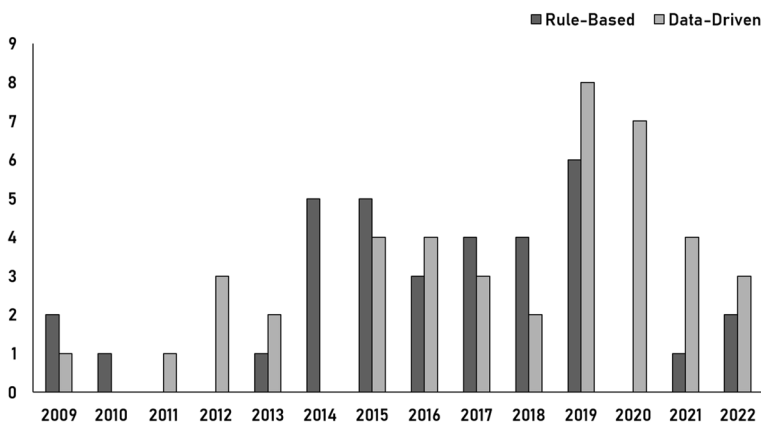


Fig. 3 Distribution of rule-based and data-driven methods per year

the distribution of rule-based and data-driven methods per year. According to Fig. 3, data-driven strategies with 42 papers were the most frequently used. On the other hand, rule-based approaches with 34 articles have been less often reported. Moreover, there has been a rise in the use of machine-learning and player modeling techniques to individualize games since 2011 and the number has reached its peak in 2019.

3.2.1 Machine-learning techniques

According to the literature, various types of machine-learning techniques have been utilized in games to formulate the DDA problem. These techniques are summarized in Fig. 2.

Search algorithms The Minimax algorithm associates a value to each state of a game and indicates how good it would be to choose an action in that state against the player. For instance, Tan used a minimax algorithm in a board game to create an AI opponent that can change an offensive factor by analyzing and evaluating players' movements to make the game harder [15]. Monte Carlo Tree Search (MCTS) is a heuristic search procedure and has been effectively applied to various games. MCTS has been used to design an AI agent for learning an action selection policy that makes the game more challenging by controlling the opponent's difficulty level [7, 39, 56, 64].

Control theory Control systems can be ideal for adapting games in response to players' states. These systems can be built based on the concepts of control theory approaches such as the classical control theory laws or fuzzy logic controls.

Parnandi described a study that provides a Proportional-Integral-Derivative (PID) controller as a mathematical tool to keep players' arousal levels around the desired setpoint for game difficulty adaptation [21].

Since player experience is a subjective factor, fuzzy logic is one of the possible ways to handle this subjectiveness issue. It should be noted that fuzzy-based methods can be classified as rule-based approaches because they make decisions based on a set of rules. However, we classified articles that employ the fuzzy method for difficulty modification as data-driven approaches. For example, Lara-Alvarez designed a fuzzy controller that analyzes users' emotional states through voice analysis and scores obtained by the players to generate the next-level aesthetic and difficulty factor [10]. This study required speech data to train an emotion recognition model. As another example, Araujo combined the advantages of neural networks and fuzzy inference control logic to train a model. This model takes player performance as input and returns the time and the number of stimuli that should appear as output in a working memory game [50]. Because the rules are derived from data and neural networks, this study can be categorized as a data-driven approach.

Probabilistic models Hidden Markov Models (HMM) are probabilistic graphical models that have been validated as a powerful tool for temporal pattern recognition. HMM can be employed to learn players' styles or predict players' performance to make an adaptive game. Zamith applied HMM to know the play style and adjust the enemy's defense strategy [45]. Enemies compose the hidden part of HMM, and players' movements are another stochastic process that produces an observable sequence.

Bayesian networks are another probabilistic graphical model that can manipulate game difficulty while keeping players engaged. Hooshyar described a study designing an

adaptive game in which a Bayesian network changes difficulty levels and provides feedback to the player for improving computational thinking in curricula [41].

A probabilistic graph is used by Xue to model and trace players' progression in a game [70]. With the help of this progression model and dynamic programming, difficulty adjustment aims to maximize the total number of rounds played, assumed to be a measure of player engagement.

Another study uses N-gram models, a probabilistic model, to estimate the probability of quitting a game [77]. Their proposed system changed the game environment to minimize players' likelihood of leaving the game.

Reinforcement learning (RL) Due to the uncertain nature of players' behaviors, a real-time difficulty adjustment method needs to make personalized adaptation decisions in a stochastic environment.

RL is a class of machine-learning techniques concerned with making decisions in an uncertain situation. In the RL process, an agent selects an action in each state to maximize the total reward. In the context of DDA, an RL agent's action is to change game parameters. By observing the outcomes of changing these parameters on the player state, an RL agent learns to make decisions that maximize players' performance or enjoyment.

According to the literature, several policies were employed to model an RL agent's behavior. For example, Zhang considered a Multi-Armed Bandit (MAB) and a novel Bootstrapped Policy Gradient (BPG) framework to formulate difficulty adjustment issues as an RL problem [42]. It also provides a clustering-based approach for learning a personalized difficulty ranking to capture users' differences. In another study, da Cruz designed a serious game in which adaptive difficulty is implemented through Q-learning [44]. The states are determined based on the game element's velocity, and three possible actions that change the game's difficulty are increasing, decreasing, and keeping up the speed. In addition, Pagalyte focused on implementing the SARSA algorithm to learn an agent that balanced battle in a fighting game by choosing appropriate actions based on players' skill levels [47].

Regression models It is possible to consider the DDA problem as an optimization problem [26]. For instance, a regression model is constructed to investigate the relationship between the five personality traits [101] with player engagement and gameplay duration [14]. Afterward, the proposed model selects the appropriate difficulty adjustment method that maximizes engagement and gameplay duration. In another study, logistic regression was utilized to estimate the difficulty value and players' failure probability [59].

Neural networks (NNs) NNs are potent tools for mapping game data to players' emotions and predicting players' experiences. Shahzad Farooq adopted the combination of deep convolutional and Long Short-Term Memory (LSTM) networks to classify players' sentiments and then used network output to adjust the difficulty [102]. Anxiety is detected if the network estimation is more than a threshold, and their proposed adjustment mechanism should make the game easier. Otherwise, boredom is expected, so the game should become more complex. Most studies used NN technology in the context of player modeling to exploit players' behaviors and experiences during gameplay. These approaches will be clarified in Sect. 3.2.2.

Each of the aforementioned machine-learning techniques adheres to a particular framework. For instance, to formulate a DDA problem using the reinforcement learning approach, the set of states, actions, and a reward function must be defined in accordance

with the DDA system's objective. Figure 4 depicts the components required to implement each approach, and examples of the definitions used by various research for these components. For instance, Lara-Alvarez employed a fuzzy controller for applying DDA [10]. This controller used players' emotional states and scores as input while the output was the difficulty factor for the next level. The information provided in Fig. 4 can provide researchers with a better understanding of how to implement DDA using different machine-learning techniques.

According to Table 4, several studies marked with an asterisk utilized player modeling to tailor game difficulty to different players. These studies' "method" column indicates the player modeling technique.

3.2.2 Player modeling

Player modeling, which has received a lot of attention in the game area, is a study that aims to detect, predict, and express human behavior or emotion [29]. It involves using data about players' behavior, preferences, and performance over time to represent their

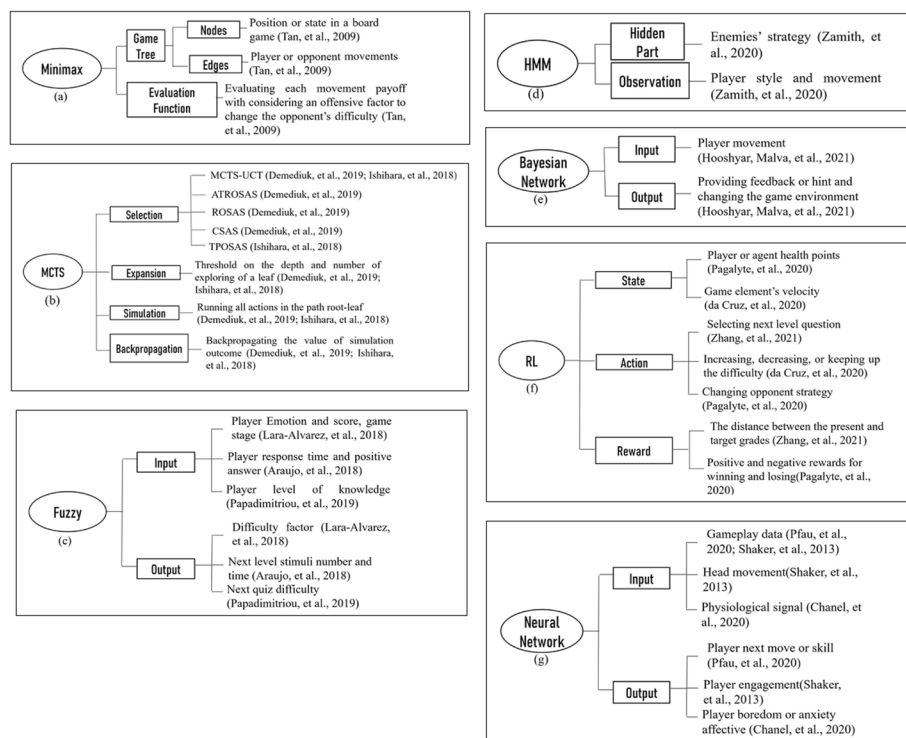


Fig. 4 Machine-learning techniques used in DDA. Ellipses represent the technique and rectangles represent the elements needed to implement that technique. MCTS-UCT, ATROSAS, ROSAS, CSAS, and TPOSAS are abbreviations used for Monte Carlo Tree Search (MCTS) with Upper Confidence Bound 1 applied to Trees (UCT), Adaptive True Reactive Outcome Sensitive Action Selection, Reactive Outcome Sensitive Action Selection, Challenge Sensitive Action Selection, True Proactive Outcome-Sensitive Action Selection, respectively

skills, knowledge, and other characteristics in a digital format. The player is an important aspect of an adaptive game. Indeed, DDA needs to consider a description of players' states, which can be inferred from gameplay data or psychophysiological signals. Consequently, researching player models to investigate gamers' preferences and integrating them with DDA approaches is a promising area for promoting engagement in games [26]. As depicted in Table 4, player modeling approaches that are used in studies to facilitate the DDA process can be split into two categories:

Player Profiling (PP) is conducted by illustrating players' typologies and playing styles, relying on personality profiling questionnaires such as Bartle's [103] questionnaire classification or BrainHex [104] typology.

A player profile consists of static information about players' personalities and general demographic information such as age and gender that could lead to adapting game features to players. For instance, Lavoué proposed a linear model using tensor factorization to estimate players' preferences and adapt gamification features to players' types obtained from BrainHex typology [5]. In another study, Hawkins demonstrates how the game's difficulty can be dynamically adjusted based on individuals' risk profiles, indicating their degree of risk aversion or risk-taking tendencies [22]. According to the findings, those who embrace risk are likely to respond more swiftly but with lower accuracy, whereas individuals who approach cautiously may answer more slowly, exhibiting a preference for gathering additional evidence before responding. As a consequence, by discerning players' risk inclination, the game's difficulty can be finely tuned to enhance players' engagement.

In addition, some studies used personality questionnaires to model extroverted or introverted [49], and personalities based on the Myers-Briggs Type Indicator [93]. By incorporating players' traits, preferences, or characteristics into the DDA process, game difficulty can be dynamically adjusted to suit individual players and enhance their gameplay experience. Therefore, we believe that discussing player profiling in the context of DDA is relevant and valuable in understanding the various approaches used in the field.

Player Experience Modeling (PEM) indicates how a player feels while playing a game and can be inferred by analyzing gameplay data patterns or players' physiological data. The former refers to the methods that use players' behavioral input, such as performance and game-specific elements derived from the interaction between players and games (e.g. [80]). The latter, which relies on players' affective responses reflected by physiological data and body movements, is also called emotion modeling (e.g. [6]).

In addition, PEM can employ a combination of gameplay and emotional data (e.g. [66]). Many machine-learning algorithms, including regression, classification, clustering, and evolutionary learning, have been used to construct player models. Available alternatives are composed of deep neural networks [46], random forest [6], singular value decomposition [92], and neuroevolutionary preference learning [91].

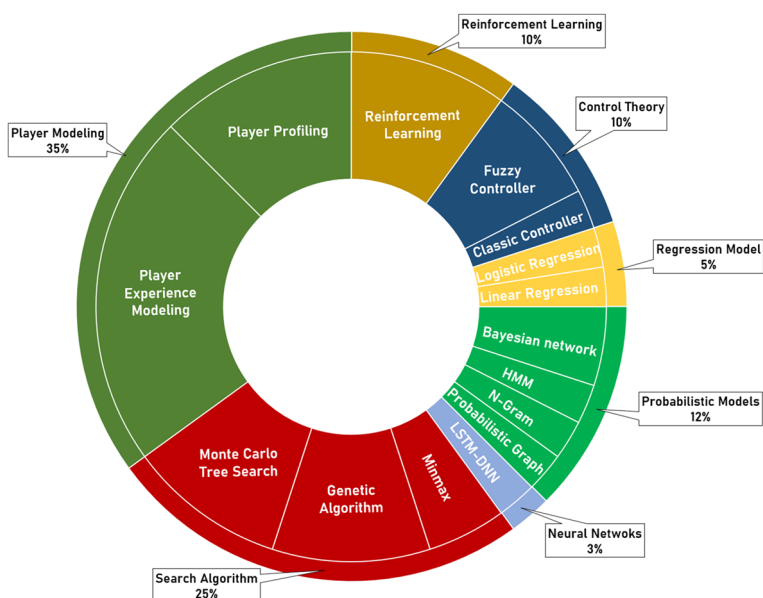
Of the selected articles, 76 developed difficulty adjustment methods (Table 4). 9 articles do not fall directly into the difficulty adjustment category. Still, they explored a player modeling approach and mentioned that the proposed method could be used for difficulty adjustment or game personalization. These player modeling articles are included in this study because of the importance of player modeling in difficulty adjustment methods. Information about these articles is summarized in Table 5. The "method" column shows the machine-learning method used to model or cluster players. Among these articles, only

Table 5 Details of studies that focused just on building a player model

Title	Player Modeling Category	Method	Game Purpose	Game Genre	Sample Size
[102]	PEM	LSTM-DNN	EG	Puzzle	50
[105]	PEM	Deep Neural Network	EG	Fighting	312
[106]	PEM	Hidden Markov Model	CRG	Puzzle	262
[23]	PP	Decision Tree	EG	Shooting	43
[107]	PEM	Deep Neural Network	EG	Role-Playing	24
[108]	PEM	Random Forests	EG	Shooting	40
[109]	PP	Hidden Markov Model	EG	Role-Playing	66
[110]	PP	Sequential Prediction	GBL	Simulation	75
[111]	PP	Linear Discriminant Analysis (LDA)	EG	Shooting	32

one study utilizes physiological data for predicting players' affective states [102]. Other studies have all used gameplay data for player modeling. The studies' objectives, important findings, and conclusions will all be included in Appendix 2.

Based on the literature analysis, the graphical representation in Fig. 5 provides a visual summary of the prevalence of various data-driven approaches. It clearly illustrates that player modeling techniques are the most commonly used methods for DDA systems. Search algorithms such as genetic algorithms and Monte Carlo Tree Search methods also represented a substantial portion of approaches. Probabilistic models, control theory, and reinforcement learning techniques each contribute shares as well.

**Fig. 5** The prevalence of various data-driven approaches

3.3 RQ-2: What types of data are employed for assessing players' states, such as performance, emotion, or personality?

The DDA system adjusts game difficulty based on an indicator of players' states, encompassing measures of their experiences or preferences. This indicator considers variables such as players' performance, emotions, and other relevant factors, providing insights into players' challenge levels and engagement. Additionally, the indicator incorporates players' preferences, including desired difficulty level and preferred gameplay features. By considering these factors, the DDA system personalizes game difficulty, creating a tailored experience for each player.

Two main data types are gameplay data and psychophysiological data. Gameplay data can be derived from players' inputs and game outputs such as players' performance (e.g. [112]), keyboard and mouse movement events (e.g., [108]), and the Spatio-temporal pattern of players' interaction (e.g., [66]). Gameplay data are computationally efficient, and no special tools are required to collect this data [113]. On the other hand, psychophysiological data is either subjective, retrieved through questionnaires (e.g., [62]) or objective, measured through various biological signals to express various players' states, such as affective states (e.g., [95]) or can be derived based on vision or voice recognition technologies (e.g., [10]).

As shown in the “psychophysiological data source” column of Table 4, biological signals are obtained through Electrocardiography (ECG), Galvanic Skin Response (GSR), skin conductance, Electroencephalography (EEG), Electrodermal Activity (EDA), and respiration. In addition, several studies also track players' head movements, eye gaze, attention, speech, and facial expressions for inferring players' states, such as affective states.

3.4 RQ-3: What are the purpose and genre of the games that DDA has been implemented in?

The purposes of video games are categorized according to whether a game is designed to provide only pleasure to players without pursuing purposeful learning, i.e., *entertainment games*, or created as a serious game to develop different skills or improve cognitive abilities [114].

Serious games used in the selected studies can be divided into three categories according to their application: *Game-based learning* are games explicitly designed for educational purposes, *Cognitive rehabilitation games* are games that help to strengthen specific areas of cognition, and *other disorder* categories are games developed to cure a specific disorder.

Figure 6 shows the purpose and genre of games used in studies to implement DDA systems. According to Fig. 6, DDA systems are primarily implemented in entertainment games. In the current review, 53 papers implemented DDA in entertainment games. 15 articles look at the role of DDA in educational games and game-based learning approaches. These games were developed to train executive function [52] and programming skills (e.g., [51]) or enhance mathematics (e.g., [20]) and computational thinking (e.g., [43]).

Furthermore, 6 studies applied DDA in cognitive rehabilitation games to improve working memory (e.g., [17]) and eye-hand coordination [44]. Eventually, 2 studies incorporated DDA in serious games for disorders such as obstructive sleep apnea [4] and amblyopia [85].

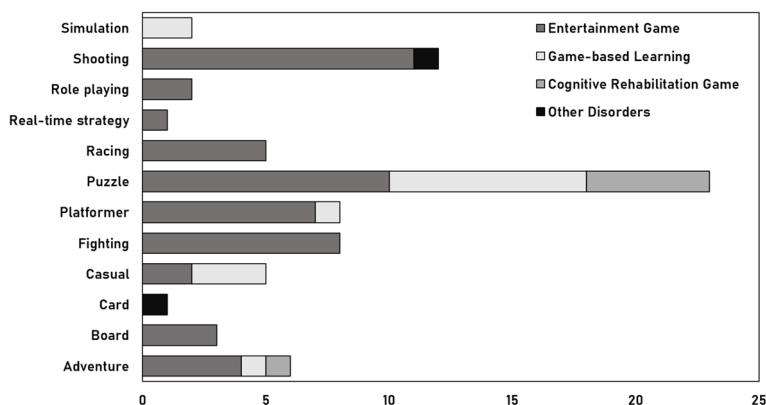


Fig. 6 The genre and purpose of tested games

Figure 6 shows that DDA has been implemented in different game genres. The puzzle game genre by far has been the most popular (23), followed by shooting games (12), platformer (8), fighting (8), and adventure (6) games. The rest of the genres accounted for less than five studies.

3.5 RQ-4: How do studies evaluate their proposed DDA techniques and identify the impact of implementing adaptive difficulty on different aspects of players' experiences?

Most of the studies provided comparative evaluation in which either compared the effects of the proposed adaptive method to a non-adaptive version or assessed the impacts of using different difficulty adjustment methods.

Several studies have used various standard surveys and questionnaires, including the Game Experience Questionnaire (GEQ) [4, 47], System Usability Scale (SUS) [44], Situational Motivation Scale (SIMS) [5], Games Engagement Questionnaire (GEQ) [77], Immersive Experience Questionnaire (IEQ) [81], Intrinsic Motivation Inventory (IMI) [37], SHORT Flow State Scale (S-FSS) [38] or other custom questionnaires to examine the effects of implementing DDA on players' experiences [8, 17].

Most studies have examined the significant effect of their proposed method on parameters such as enjoyment, flow, motivation, engagement, and immersion. For example, Zamith used HMM to adjust the game difficulty in a shooting game [45]. They compared player experience between the two versions with and without difficulty adjustment and showed that 90% of the participants found the adaptive version more fun than the non-adaptive one. In a study conducted by Denisova, the difficulty level of a shooting game was changed by modifying the game timer [81]. Results showed higher immersion when playing the game with an altered timer compared to the standard timer (p -value < 0.05). All these studies used questionnaires to demonstrate the effectiveness of their proposed DDA approach.

On the other hand, while some researchers employed questionnaires to assess subjective reporting of players' experiences, others evaluated their proposed method's efficiency through analysis of players' performance in games. For instance, Scirea showed whether playing the game with DDA significantly impacts players' progression [43], and

Jácome-Amores showed improvement in achievement level [20]. Xue used win rate as a measure for their game difficulty adjustment. They showed improved player engagement when using the with-DDA version compared to the game without DDA ($p\text{-value} < 0.05$) [70].

The efficiency of using DDA through player engagement and game achievement falls under the abovementioned areas. On the other hand, there are limited studies that have measured the impact of using DDA on improving ability as a measure of the effectiveness of DDA. For instance, Plass used a one-tailed paired t-test ($t = 1.86$, $p = 0.033$) to investigate the influence of adaptive difficulty adjustment on the enhancement of executive function abilities using a game-based intervention [52]. As another example, Correa used DDA to change the complexity of a game to get a balanced level with therapy protocols [85]. They showed that using an adaptive system can cause 2–3 levels of improvement in visual acuity using the Snellen Vision Test.

An essential factor in a human-based study is the number of subjects used. In our review, we have single-subject studies with as low as 3 subjects up to large sample size studies with 794 subjects (Table 4).

4 Discussion

4.1 The literature dissemination

Across the selected articles, there were 39 journal and 47 conference articles. It should be noted that the number of research dealing with game adaptation and difficulty adjustment is much higher than the number of papers listed in Table 4. This is due to our inclusion criteria, in which we have limited our analysis to articles involving human participants.

Additionally, we analyzed citation counts for the reviewed publications to better show the scholarly influence in the field. On average, publications received over 35 citations, highlighting the recognized impact of research in DDA.

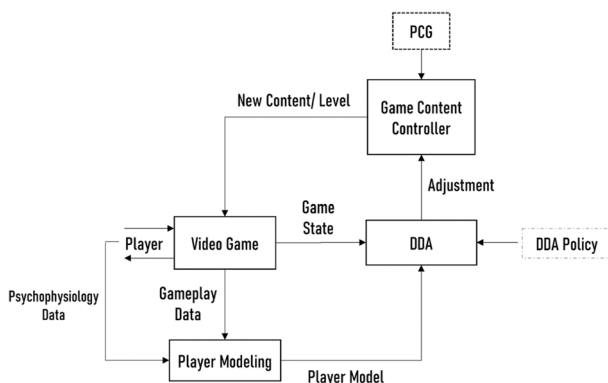
The growing trend of publication numbers through 2019 and the increased usage of machine-learning methods and player modeling since 2011 (Fig. 3) indicates that there is still interest in the DDA research field. According to the findings presented in this review, it is interesting to mention that Bayesian approaches, reinforcement learning, and fuzzy logic have garnered greater attention in recent years. These three approaches have been introduced in high-impact journals and have received an average of 39 citations in the last two years.

4.2 The structure of a video game with DDA

According to the categorization presented in this study, the structure of a video game, with emphasis on DDA, can be summarized as shown in Fig. 7. According to Fig. 7, the implementation of an individualized difficulty level in DDA can be facilitated by leveraging three key sources of information: the game state, the player model, and the DDA policy.

These sources contribute to the overall process of tailoring the difficulty level for each player, with each source providing valuable insights and inputs. In short, the game state shows the current difficulty level and the game setup that may be changed in the next step. For example, it can be the number of enemies a player should kill in each level of a shooter game [16]. The player model is a description of the player state which is required to make a

Fig. 7 Diagram of a video game with DDA



game difficulty level individualized. As explained in Sect. 3.3, the input to the player modeling module can be as simple as the raw in-game performance data or more sophisticated psychophysiological data acquired through tools such as eye-tracker, voice recorder, or questionnaire. Finally, the DDA policy is defined according to game designers' goals. For example, the purpose of the DDA may be to balance the scores of an opponent and a player in a competitive game, meaning that it tries to keep both at the same level and design a fair opponent for the player. Alternatively, the DDA policy may make the game more difficult as the player progresses.

DDA module can be developed based on any difficulty adjustment methods described in Sect. 3.1.1. Accordingly, DDA analyzes games' states and, based on current players' states, selects appropriate adjustments for the next stages. The game content controller generates new content and presents it to the player according to the adjusted difficulty level.

Another module involved in this process is the Procedural Content Generation (PCG) module, which helps generate content automatically and, if necessary, personalized. For example, suppose the DDA module is implemented with the help of reinforcement learning for a shooting game. In that case, the number of enemies and the player's score determine the game state and the player state, respectively. Then, the reward function is defined based on the DDA policy (e.g., keeping players' scores within a specific range). The output of the DDA module finally determines the appropriate difficulty level for the player.

Considering the structure illustrated for a DDA system in Fig. 7, difficulty adjustment is a design challenge that requires predicting when and how to interfere. It is incredibly challenging to keep the players engaged in interactive environments. The challenges and requirements of a DDA system can be outlined as follows according to the different dimensions shown in Fig. 7.

4.3 Analysis of different aspects of DDA

As illustrated in Fig. 7, the first step in designing an effective DDA system is to closely examine the gameplay and level design to determine the key variables that influence difficulty and can be adjusted in real-time. This process applies to every game genre and entails recognizing gameplay elements such as time limits, environment map layouts, enemy strength, and numbers that shape the challenge presented to players [13]. Once these adjustable parameters are determined, the subsequent challenge lies in appropriately

fine-tuning them according to the mechanics specific to each genre, as each genre manifests difficulty through distinct means [24].

For instance, in puzzle games, important parameters often include the number of moves required [42, 44], the availability of hints [112], and the complexity of spatial reasoning needed [38]. Shooter games, on the other hand, focus on factors like enemy speed [45], utility [55], level chaos [93], and introducing randomness into enemy behaviors [115]. In Role-Playing Games (RPGs) difficulty can be adjusted by controlling the pace of advancement by linking it to enemy scaling [3]. Platformers adjust the difficulty by expanding the number of obstacles, the speed of platform movements [62], fine-tuning jump precision [67], and presenting challenges that demand quick reactions [16]. Fighting games incorporate the skill level and strategies employed by AI agents [105], and intricate move sets [47]. Racing games adjust difficulty by modifying AI opponent driving experience [73] or track complexity [66]. Many parameters of both the enemies and the player character itself can be adjusted to change the difficulty level in action-adventure games [37].

As each genre poses diverse challenges to players, the set of parameters that should be modeled and controlled is largely unique and depends on the core mechanics and skills associated with the genre. Game parameters should be sensibly chosen to align with the aspects that make each genre inherently difficult. Therefore, a single universal definition of game difficulty that applies across all genres does not exist [24].

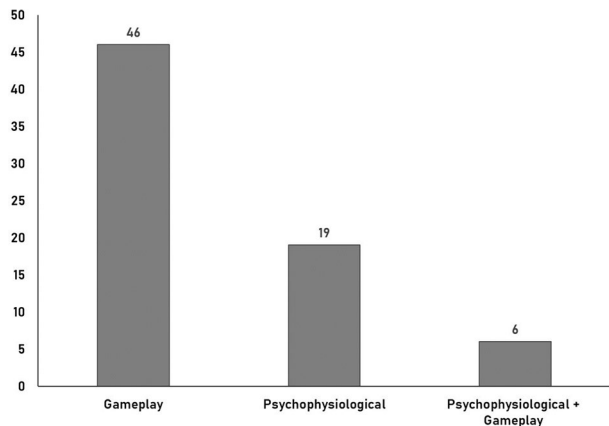
In line with the aims and research questions mentioned in this paper, the following discussion and analysis will center on previous studies connected to DDA. This examination will analyze the data sources, adjustment methods, game genres, and evaluation methods used in previous studies on DDA to demonstrate its impact on players' experiences.

4.3.1 Data

DDA requires data in order to determine games' level of difficulty. DDA systems can make decisions based on a variety of data types, each with its own set of benefits and drawbacks. In the present study, the data types employed for assessing players' experiences are classified into the following categories: gameplay and psychophysiological data. Figure 8 shows the number of papers in each of these categories.

As shown in Fig. 8, the majority of the included studies opted to adjust the difficulty using gameplay data. This finding aligns with Paraschos's study, which also reported a

Fig. 8 The number of papers focusing on each data type category



predominant use of in-game data for player assessment, facilitating the customization of a game environment to suit individual players' skill levels [25]. Designing a DDA system based on gameplay data is computationally efficient, is the least intrusive for players, and is cost-effective [113]. However, when these measurements are the only source of information, they cannot provide enough information about individual players, which may result in inferring incorrect players' states [116]. For instance, considering players' experiences, including being an expert or a casual player, significantly impacts player enjoyment [89].

Furthermore, certain studies leverage gameplay data to construct models of players' profiles [22] or experiences [105]. While these data offer efficiency, they may lead to low-resolution models, particularly when it comes to capturing affective components [117]. These models often rely on several strong assumptions that attempt to establish connections between players' experiences, gameplay actions, and preferences [22]. Merging gameplay and psychophysiological data and considering multimodal player models can offer more potential models to explain personal experiences and preferences [91].

Psychophysiological data provide players' feedback either directly through players' self-reports or indirectly through biological signals, vision, or voice information [117]. Each method provides distinct advantages as well as potential limitations. Self-report techniques like surveys allow researchers to directly access players' thoughts, feelings, and perspectives on games [118]. Players can convey both their cognitive states, such as mental challenge or engagement [72], and emotional reactions like excitement or frustration [62]. Self-report techniques are scalable and provide the ability to gather data from a large number of players [119]. However, self-reporting can be impacted by response and recall biases. When using self-reporting questionnaires to gather information about players' feelings and experiences during gameplay there are possibilities of memory bias. Since players answer questions after playing games, their responses might be influenced by their memory of the gaming experience, rather than an accurate representation of their feelings at the moment [117]. To mitigate this issue, researchers can employ other data collection methods, such as physiological measurements (e.g., skin conductance and heart rate), to complement self-report data [120]. Combining multiple data sources can offer a more comprehensive and robust understanding of players' experiences [91, 117].

On the other hand, biological signals, such as EEG, skin conductance, and heart rate can offer objective indicators of players' physiological states and emotional responses [67]. Indeed, it is true that these sensors can be costly, and their primary application, as suggested by Bontchev [2], still remains largely limited to academic and simulation experiments rather than commercial games.

While methods like analyzing facial expressions [6] and speech [10] may be technically possible in games, there are questions about how useful facial and speech analysis actually are for recognizing emotion in games. This is because most players tend to sit still and stay silent during gameplay, providing limited facial and vocal cues to interpret [117].

In addition, a DDA system should not require a lot of data to calibrate the game. This can cause the "cold-start" problem [70]. As supervised and unsupervised machine-learning suffers from the "cold-start" problem, in the matter of game adaptation, this problem is raised when the gameplay data is unavailable or first confronted by a new player. The simple solution to this problem is to allow players to play the initial levels without considering any adaptation mechanism. The level adjustment could be applied to the next levels based on the collected data. In this case, the problem of properly initializing difficulty parameters exists. Another solution is to use transfer learning concepts that use the knowledge about players' experiences obtained in one domain for prediction in another [102].

4.3.2 Adjustment method

According to Fig. 7, in the second step, the DDA mechanism adjusts the difficulty based on the available information about players' states using one of the techniques described in Sect. 3.1.1. The methods used in the included articles are further evaluated in the following.

As illustrated in Fig. 6, DDA approaches have been examined in games with diverse purposes, genres, and persons. As a result, there is no one criterion by which they can be compared [27]. In other words, there is no benchmark for applying a uniform comparison between different DDA methods. Most articles that compared their proposed system to a non-adaptive one claimed that utilizing a DDA system improves players' experiences. However, there is insufficient evidence to determine the best technique used in DDA.

Each of the approaches listed in Sect. 3.1.1 has advantages and downsides, and some may only be utilized for a specific genre. For instance, Minimax is mostly used in board games and tries to build a full game tree [54]. It also requires an evaluation function to calculate an estimate of games' outcomes in each state. On the other hand, this technique ignores players' experiences and levels because it always assumes that players play perfectly, i.e., players make the best move at each step of the game [15]. This method is not suitable for complex games where players' movements cannot be restricted, and the whole game tree cannot be created [56]. The advantage of this method is that board game analyses will be of the same shape and will be limited to tree structure analysis [24].

Monte Carlo Tree Search (MCTS) techniques were employed in games with large state space. Unlike the Minimax method, MCTS-based methods do not entirely explore a sub-tree and, due to their reliance on randomness, use the average of such random samples as an estimate for the value of the entire sub-tree [56]. This reliance on randomness has the advantage of not requiring defining an evaluation function. However, like the Minimax method, this one is more commonly employed in competitive games (e.g., fighting games) to determine opponents' levels of difficulty [64].

Due to the large state space in several games, the genetic algorithm is another method focused on searching and finding the desired level. It is commonly used in the PCG field since it provides highly customized content that evolves the levels according to players' progress [4]. In this algorithm, the population may be defined according to the game-level parameters. Genetic algorithms have been primarily employed in games with the puzzle and platformer genres [16, 76].

The fuzzy algorithms have been employed in the DDA studies to better handle vagueness in the description or interpretation of difficulty [51]. As shown in Table 4, this strategy may be applied to various genres, including puzzles [10], adventures [51], and racing [21]. In general, it can be used to map real values to output parameters presented by fuzzy rules. The hardship with the fuzzy method is that the definition of membership functions and rule sets is quite specific and requires expert knowledge [121].

Neural networks can be used as an approximation function for mapping between player and game state [105] or as a controller for difficulty adjustment [48]. According to Table 4, most studies used NN technology in the context of player modeling to exploit players' behaviors and experiences during gameplay.

Despite the exploitation of varied machine-learning approaches, it is still impossible to provide a definitive and precise solution to the DDA issue. The implemented methods

are highly dependent on the type and genre of the game [27]. For example, reinforcement learning methods have been further used in competitive matches to balance the abilities of players and NPCs [47]. On the other hand, genetic algorithms are primarily used in cases where content generation is also intended in addition to adjusting the difficulty [122]. In addition, adaptive procedures that change the difficulty only based on the most recent performance may offer an inappropriate level to a player [106]. Hence DDA approaches need to track players' progressions during gameplay. Thus, researchers have further employed HMM and LSTM for this purpose [45, 48].

Along with all of these, one of the basic requirements of a DDA system is that the game difficulty must be changed in real-time [27] and in a way that players do not notice. Awareness of DDA presence can negatively impact player performance and reduce DDA effectiveness [86].

4.3.3 Game genres and the effect of DDA

The genre of a game plays a key role in determining the difficulty parameters, adjustment methods, and overall DDA implementation. The articles reviewed in this study cover a diverse range of genres, including puzzles, shooting, fighting, adventures, and more. Most studies have examined DDA in single-player games, while only 8 articles have explored DDA in multiplayer games, with a focus on racing titles. Designing effective DDA systems for multiplayer games poses unique challenges [123]. As explained by Baldwin, while existing DDA research focused on adjusting difficulty in real-time for single-player games, little explored applying DDA in competitive multiplayer contexts where challenge stems from human opponent skill levels [124]. Implementing DDA in multiplayer games presents distinct challenges versus single-player experiences. One challenge is balancing difficulty across players with varied skill levels within the same match [65]. If the DDA system fails to assess and adjust players' abilities accurately, it could negatively impact players' experiences. For example, consistently matching a new player against very skilled opponents could become frustrating and cause them to quit [87]. Single-player DDA adapts to one player, while multiplayer DDA must account for experts and novices simultaneously [73]. Another key challenge is maintaining fair competition between players of different abilities [87]. Implementing fair and balanced DDA in multiplayer games remains a complex challenge requiring further research [73]. As Baldwin discussed, additional studies are needed to better understand players' experiences, identify the most effective DDA components for various players, and expand on the preliminary framework proposed [124]. Given the multifaceted nature of successful multiplayer DDA design, a comprehensive examination exceeds the scope of this current review. However, the questions raised highlight fruitful directions for future investigation in this emerging area.

The reviewed articles pursue DDA in serious games or entertainment titles. Across both categories, studies demonstrate that DDA can enhance motivation [5], immersion [81], engagement [70], enjoyment [56], and performance [13]. It can also improve abilities like executive function [52], computational thinking [41], programming skills [51], and visual acuity [85]. This review focuses on papers that test DDA methods on human participants, yielding more accurate and reliable results on players' experiences than simulations alone [25]. However, some studies use relatively small sample sizes. For example, Scirea [43] reports improved progression and computational thinking among only 3 participants when testing a DDA-based puzzle game. With such a small sample, the results have limited generalizability. In contrast, another study by Hooshyar on an

adaptive computational thinking game involved 79 subjects, lending more credibility to the findings [41]. Overall, human-centered DDA evaluations provide important insights, but larger participant samples are needed to further validate effectiveness and derive more authoritative conclusions.

In addition to enhancing engagement and flow for players, adaptive difficulty serves an important role in making games more accessible to diverse populations of players [49]. Dynamic adjustment of in-game challenges can help accommodate players who have disabilities or impairments that affect their gameplay abilities [40]. By tuning the difficulty to match individual skill levels, games can provide more equitable access for players with sensory, motor, or cognitive differences.

Player modeling techniques, including both player profiling and modeling of individual cognitive limitations, present promising opportunities to enhance DDA to make games more accessible. Considering the human cognitive constraints and features such as working memory capacity, attention, response time, etc. can help to understand their preferences better and, as a result, better design adaptive games [125]. Limited studies have implicated human biological constraints in designing game difficulty levels. For example, Koulouris proposed a tool that estimates the probabilities of objects to be attended to in real-time by considering human visual features and then used this tool to adjust game difficulty levels by objects' repositioning [78]. Considering players' cognitive characteristics and biological constraints, particularly in serious games, may open up new avenues for DDA research. For instance, finding or detecting a target is essential to many video game genres. The visual attention of players is important in this kind of game. As a result, modeling players' cognitive features can help better design games' difficulty levels.

In addition, by maintaining detailed player profiles, DDA systems can closely tailor difficulty to an individual's strengths, weaknesses, preferences, and specific accessibility needs related to physical, sensory, or cognitive impairments [94].

As depicted in Fig. 7, game designers can define a DDA policy. This policy could be based on the principle of fair play, aiming to strike a balance between players' abilities and challenges posed by opponents, or it could be tailored to increase games' difficulties based on players' progress. However, it is important to recognize that such policies can have varying effects on different individuals [14]. Player types differ in their motivations and preferences, as exemplified by Bartle's player types categorization [126]. For instance, in this typology, achievers, who find satisfaction in progressing, gaining power, and honing their skills, might find DDA that constantly matches enemies to players' levels, less enjoyable [23]. Thus, it is crucial to take into account the preferences and motivations of different player types in the design of DDA systems to ensure a diverse and enjoyable gameplay experience.

While detailed player profiles and consideration of player types contribute to tailoring DDA, certain limitations persist in fully understanding how DDA can effectively support diverse audiences. Notably, a significant gap exists in the exploration of how DDA may differentially impact players from various cultural backgrounds, indicating the need for future research to uncover the cultural influences on players' experiences.

Moreover, there are not many studies addressing the effect of DDA on different age groups. There is a study focused on adaptive versus non-adaptive gameplay in a specific serious game for executive function improvement among players aged 10–17 [52]. Results showed age-dependent impacts, with older participants facing greater challenges. However, as the study is confined to a narrow age range and educational context, drawing definitive conclusions about DDA's influences on learning outcomes or engagement for diverse age groups remains unknown.

Furthermore, there is an opportunity for a more in-depth examination of how DDA outcomes may vary based on players' skill levels or expertise. Certain studies underscored that novice players often prefer easier challenges, while more experienced gamers seek higher levels of competition [89]. However, our analysis revealed limited research directly examining how DDA affects players of varying skill levels. Few studies organized participants into discrete categories such as beginners, intermediates, and experts to analyze DDA impacts separately. While Cechanowicz demonstrated that balanced difficulty improved experiences across both novice and skilled players alike [87], Lora-Ariza reported increased flow only for less experienced groups, neglecting insights into experts [38]. Broadening the focus to encompass players' abilities and experiences collectively can result in more precise difficulty settings, enhancing the gaming experience for a wider audience [94]. More research is needed to fully comprehend DDA's influence on all players, especially highly skilled players. However, it's important to note that conducting comprehensive studies on the impact of DDA across diverse demographics, including cultural backgrounds, age groups, and skill levels, faces significant challenges due to the limited access to extensive player data. This data is often proprietary to large gaming companies and not readily shared with the broader research community.

According to Table 4, many DDA studies rely on simple, bespoke games rather than popular commercial titles. While puzzles, shooting, platformer, and fighting games are common genres examined, most racing game studies used arcade-style titles [21] and only one study implemented DDA in an advanced AAA-style racing game [66]. This indicates a gap in studying DDA in more complex action-adventure, RPGs, and strategy genres. For example, only one study analyzed a real-time strategy game [71], and none examined popular action RPGs or action-adventure games like Souls-like titles.

Electronic Arts (EA), as a leading company in the area of video games, has secured a patent for its DDA methodology and conducted testing across various level-based games, consistently yielding heightened player engagement [70]. In addition, renowned titles like *Left 4 Dead*, and *Silent Hill* have incorporated DDA, as discussed in [111]. However, examining the literature reveals that the majority of commercial games do not provide detailed insights into their DDA mechanisms [27].

The focus on simple game mechanics in academic studies provides controlled experimental conditions but may lack real-world relevance. More research is needed on implementing DDA in commercial games across a wider variety of genres, especially complex RPGs, open-world adventures, and strategy games. Studying DDA in more genres would enhance the generalizability of findings and provide practical guidance for popular titles.

While DDA is often believed to enhance players' experiences, some criticisms exist. Soderman argued that DDA is a marketing strategy aimed at prolonging playtime and increasing spending [99]. It is true that DDA systems can be utilized as a means to prolong player engagement, which, in turn, can lead to increased revenue in the gaming industry.

However, it is essential to distinguish between the primary goals of DDA in entertainment games and its application in serious games. While DDA in entertainment games may have a dual purpose of enhancing players' enjoyment and prolonging engagement for commercial reasons, the use of DDA in serious games is primarily focused on improving the learning experience and achieving specific educational or training objectives [28].

In serious games, the primary objective is to facilitate effective learning [52] or skill development [60]. DDA is employed in serious games to adapt the gameplay experience to the individual player's needs, ensuring an optimal balance between challenge and skill level. By maintaining a state of flow, serious games aim to maximize players' engagement, motivation, and gameplay duration, leading to better learning outcomes [79].

Due to the unique characteristics of each player, such as gender, skill level, personality, preferences, and moods, DDA systems face challenges in accurately tailoring difficulty levels to individual preferences. In other words, although a perfect DDA system is supposed to provide the right level of difficulty for each player, however, it is very hard, or nearly impossible, for DDA systems to consistently provide an ideal gaming experience for each player. As a result, some players may be negatively affected by inappropriate difficulty adjustments, hindering their enjoyment of the game.

As mentioned in Sect. 3.1, in contrast to DDA methods, there are player-adjusted methods in which a player directly manipulates the difficulty level by changing the game mechanics. The question of whether DDA techniques are more suitable than player-adjusted methods is yet to be answered. Ang showed that DDA could improve performance compared to player-adjusted levels [69]. This is despite the evidence that player-adjusted level results in more engagement. On the other hand, based on self-determination theory, the player-adjusted approaches allowed players to set goals for themselves, satisfying their need for autonomy and subsequently increasing their enjoyment [88].

DDA and player-adjusted approaches have their drawbacks. The DDA approaches suffer the risk of getting stuck in local extrema or may be accompanied by large fluctuations [75]. On the other hand, in the player-adjusted approach, allowing players to change the game difficulty by interacting with the game environment could distract them. Furthermore, players may not understand their game competency level correctly, resulting in frustration or boredom. As a suggestion, the optimal approach can be a combination of player-adjusted and DDA methods to make a trade-off between performance and enjoyment. Such a system would benefit from the autonomy factor in games, which is implemented by a player-adjusted approach while benefitting from DDA for better performance.

5 Conclusion

This systematic review investigated game adaptation techniques by examining how DDA can be adopted in different games. We identified 85 papers that met the inclusion criteria. This article presents a structure for the DDA module in video games, which clarifies the role of the DDA process in the game engine. In addition, various aspects of DDA techniques, including the utilized machine-learning and player modeling techniques, the data types used for assessing players' states, and the application of DDA in different game categories, are examined. This information can assist researchers in choosing an appropriate method to implement DDA in their game. Furthermore, by providing information about previous evaluation methods, researchers are better able to assess the effectiveness of their approach to players' game experiences. Results show that future games should consider individual differences through player modeling. For the sake of the complex nature of human players and to strengthen the existing DDA techniques, further research is required in this area. Modeling players' cognitive characteristics and biological constraints can be a novel way to predict players' preferences and change game difficulty levels accordingly, which has received less attention in the literature. Although there has been significant research on dynamically adjusting difficulty in real-time for single-player games, further studies are required to gain a better understanding of players' experiences in competitive multiplayer environments. Additionally, there is a gap in investigating DDA in more intricate game genres such as action-adventure, RPGs, and strategy games.

Appendix 1

Table 6 Lists of included articles that applied dynamic difficulty adjustment

Title	Year	Publication	Target	Evaluation and result
[37]	2022	J	Creating a biofeedback-controlled DDA that adapts the game's difficulty based on the player's heart rate	Conducting a One-Way ANOVA to compare the means between the experiences reported by the two groups of players, either with or without the DDA ($p < 0.05$)
[38]	2022	J	Proposing a Case-Based Reasoning approach for dynamically assessing player skill levels and adapting the game difficulty based on the last state of the game	Reporting higher flow experience with $p < 0.017$ using DDA (balanced version) than without it (normal version)
[39]	2022	J	Designing game AIs that consider player states for DDA to automatically and progressively adjust AI opponent actions	Conducting a Wilcoxon Signed Rank to show that the proposed AI opponent caused a significantly higher player experience ($p < 0.05$) than the Heuristic-based agent
[8]	2022	J	Implementing an adaptive rubber-banding AI to improve the adaptability of the DDA in a racing game	Claiming better game experience when competing against an adaptive rubber-banding AI agent
[40]	2022	J	Developing a mobile game that adapts to the level of the player to facilitate vocabulary learning for children with autism spectrum disorders (ASD)	Playing the DDA-Mode for children resulted in 88.4% higher engagement than playing the classical levels-mode
[41]	2021	J	Designing an adaptive game for promoting computational thinking (CT) in curricula in which a Bayesian network modifies the difficulty level and provides feedback to the player	Conducting an ACNOVA test with $F = 6.20$ to assess the impact of an adaptive version of the AutoThinking game compared to the traditional version on increasing Computational Thinking Skills
[41]	2021	J	Designing an adaptive game for promoting computational thinking (CT) in curricula in which a Bayesian network modifies the difficulty level and provides feedback to the player	Conducting an ACNOVA test with $F = 6.66$ to assess the effectiveness of an adaptive version of the AutoThinking game compared to the traditional version in increasing Computational Thinking Skills. Using a t-test, researchers looked into differences in students' interest ($t = 2.52$), satisfaction ($t = 2.66$), flow states ($t = 3.50$), and technology acceptance ($t = 2.00$)
[10]	2021	J	Using a fuzzy controller to change next-level aesthetic and difficulty factors based on users' emotional state (through voice analysis) induced by previous level aesthetic and player score ratio	Using the t-test, increased pleasant-high states, decreased unpleasant-low states ($t = 2.18$), and maximum Score ($t = -0.98$) were reported when comparing fuzzy DDA and Linear DDA

Table 6 (continued)

Title	Year	Publication	Target	Evaluation and result
[42]	2021	J	Developing a data-driven strategy using reinforcement learning for adaptively adjusting difficulty levels and providing a clustering-based methodology for learning a personalized difficulty ranking to reflect users' differences	In three modes, i.e., no-adaptation, rule-based adaptation, and machine-learning-based adaptation, paired t-tests were used to see if the memorization time was changed from the beginning to the end of the game. Results for fast and slow players in each mode are as follows: for fast players: no-adaptation $t = -0.45$, rule-based adaptation $t = -4.35$, machine-learning-based adaptation $t = -2.64$ for slow players: no-adaptation $t = 3.08$, rule-based adaptation $t = 7.05$, machine-learning-based adaptation $t = 11.18$
[20]	2021	C	Changing game difficulty based on the number of achieved stars by the player to improve children's math learning	Conducting a t-test with $t = 8.76$ to report the improvement in achievement levels between the game version with DDA and the normal version
[43]	2020	C	Creating an adaptive level generation mechanism that adjusts the difficulty of the next level based on the player's performance in prior games	Reporting the improvement of player progression to complete puzzles during the time (no statistical analysis reported)
[4]	2020	C	Proposing a procedural content generation technique to generate new NPC based on players' choices and their progression and also adjusting game difficulty by changing the number of NPC features based on players' win rates	Using the Game Experience Questionnaire to identify differences in user experience between three versions of the game: version A (difficult version), B (adaptive version), and C (random version). the results of conducting the t-test are as follows: group 1: when players played versions A and C, there was an increase in unpleasant experiences ($t = -3.42$) group 2: when players played versions B and C, there was a decrease in unpleasant experiences ($t = 2.53$)
[44]	2020	C	Using Q-Learning to adjust the difficulty in a serious game by increasing, decreasing, and maintaining the speed of game elements	The System Usability Scale (SUS) questionnaire was used to assess the game's usability. Given that the average acceptable score for this test is 68 points, the proposed system has gained an average of 74 points
[45]	2020	C	Using HMM to adjust the difficulty by changing enemies' defense strategy in a shooting game	A brief interview was done to compare the player's experience between the adaptive and the default version of the game. The adaptive version is preferred by 90.9 percent of participants

Table 6 (continued)

Title	Year	Publication	Target	Evaluation and result
[46]	2020	C	Using a dataset from the popular massively multiplayer online role-playing game (MMORPG) Aion, deep network prediction was used to automate game balancing by selecting the next player's skill or action	Conducting a one-way ANOVA test with $F=9.63$ to assess the significant difference in adaptive agent proficiency
[47]	2020	C	Using a reinforcement learning agent to keep battles balanced by selecting suitable actions based on players' skill levels	The Game Experience Questionnaire was used to assess the player experience in both the adaptive and non-adaptive versions
[48]	2020	C	LSTM-DNN is used to classify players' emotional boredom or worry, and the difficulty is adjusted based on the network's output. There are two DDA approaches: 1-Absolute method: if the network's output is greater than 0.5, anxiety is identified, and DDA reduces difficulty; if the output is less than 0.5, the difficulty increases 2-Relative method: evaluates network's output between t and t-1 and reduces difficulty if the output is better	Reporting the accuracy of the LSTM network (73.2) compared to QDA (66.1) and LDA (99.5). Users' perceptions were compared between relative and absolute adaptation methods using the Binomial significant test
[5]	2019	J	Adapting gamification features in a puzzle game using a player model that evaluated a feature's preference using a weight	Using the Situational Motivation Scale (SIMS) questionnaire has been compared the suggested adaptive technique to two other methods in terms of spending time, player motivation, and enjoyment. Results showed that different game elements could influence user motivation and involvement differently
[49]	2019	J	At the beginning of the game, a questionnaire was used to model player personality, and two games' environments for introvert and extrovert players were manually defined	Using a t-test to show the impact of the personalized educational game on enhancing motivation ($t=1.71$, $p<0.05$), reducing cognitive load ($t=3.64$), and receiving more positive comments on perceived utility ($t=-2.13$, $p<0.05$) when compared to a non-personalized educational game
[50]	2019	C	Training a model that takes player performance (reaction time and positive answers ratio) as input and returns the time it takes for the stimulus to appear as well as the number of stimuli as output	ANOVA test with $F=6.99$ has been shown that a higher N-back level obtained in the game version with DDA compared to the default version
[51]	2019	J	Developing a student model for adjusting adaptive features in an adventure game based on the learner's knowledge level	Usability, likeability, educational effectiveness, and adaptivity are all excellent and encouraging things to report (not statistical analysis reported)

Table 6 (continued)

Title	Year	Publication	Target	Evaluation and result
[13]	2019	J	Proposing a six-step plan for adjusting the difficulty	Evaluating performance improvement between the proposed balancing method and a version without a balancing method by applying it to two different game genres: a fighting game and a platformer game (no statistical analysis reported)
[6]	2019	J	Proposing a Facial Expression Analysis (FEA) as a model for changing game difficulty based on a heuristic method, as well as a tool for evaluating perceived in-game difficulty by training multiple classifiers	T-test results have shown that participants prefer the personalized adaptive version to the static one in three game modes: easy mode: $p=0.0001$, normal mode: $p=0.0008$, hard mode: $p=0.005$
[52]	2019	J	Comparing the effectiveness of an adaptive vs. a non-adaptive game for improving executive function skills	A one-tailed paired t-test with $t=1.86$ was used to investigate the influence of an adaptive difficulty adjustment on the strengthening of executive function skills
[53]	2019	C	Generating and adjusting game levels using predefined rules and constraints based on capturing player actions during gameplay	The proposed level generator is preferred by participants over existing level generators (no statistical analysis reported)
[54]	2019	C	Proposing a DDA technique in which a finite state machine agent with multiple difficulties is created, with a heuristic function that switches between them according to the performance of the players	A 74% win rate revealed that the difficulty level assigned by the proposed DDA method is adequate, and the flow channel was achieved in 79% of the participants
[55]	2019	C	Using obtained provenance data to dynamically tune the game's challenge, which means that game elements, players, and actions were assumed to be an entity, agent, and activity, respectively, and the difficulty was altered based on the relationship between these parts	When compared to the non-DDA game version, the proposed DDA approach provided more opportunities for participants with lower skills and reduced the average survival time (no statistical analysis reported)
[56]	2019	C	Comparing 4 DDA agents in a fighting game: 1—Monte Carlo Tree Search (MCTS) agent, 2—Challenge Sensitive Action Selection (CSAS) agent, 3—Reactive Outcome Sensitive Action Selection (ROSAS) agent, 4—Adaptive True Reactive Outcome Sensitive Action Selection (ATROSAS) agent	Win percentages of each of the agents were reported: MCTS = 90.323, CSAS = 40.860, ROSAS = 46.335, ATROSAS = 51.075
[57]	2019	C	Investigating the influence of a DDA system on players' perceptions of their chances of success and players' confidence compared to a random difficulty curve	Based on questionnaire responses, the ROSAS agent was the most enjoyable (mean response of ≈ 2.4), and the MCTS agent was also the least enjoyable (mean response of ≈ 2.4)
[58]	2019	C	Clustering players into three clusters: newbies, average, and experts based on data captured from 10 previous movements and changing game difficulty based on several rules	The findings demonstrate that DDA systems can cause players to become overconfident Comparing the improvement of player's enjoyment between an adaptive and non-adaptive version of the game (this improvement is not statistically significant)

Table 6 (continued)

Title	Year	Publication	Target	Evaluation and result
[59]	2019	C	The initial levels' difficulty was adjusted based on interpolation between the easiest and hardest fixed values for the difficulty that the designer defined. Then, logistic regression was used to estimate the difficulty value and players' failure probability	Results showed that the model is quick enough to allow players to discover the gameplay with an average convergence time of 105 s. The model's failure prediction accuracy was reported at 0.82
[60]	2018	J	Using two questionnaires to model play styles and learning styles, manually designing the game and learning contents depending on trained models	When students played a personalized version of the game, the results revealed a positive engagement
[61]	2018	C	Creating a game to help students improve their math skills, then comparing two versions of the game: adaptive and non-adaptive. The next-level difficulty provided to a player in the adaptive version is determined by how this player has resolved the current level	Expressing interest in using the developed game in an educational environment through a questionnaire surveying educators' thoughts (no statistical analysis reported)
[62]	2018	C	Recognizing players' emotions by analyzing a textual conversation between a player and an NPC and adjusting the difficulty based on several rules	Compared to static difficulty adjustment, the proposed DDA provides a better player experience with significantly higher scores ($p < 0.005$)
[63]	2018	C	Face recognition is used to detect the emotion of players, and then the difficulty is adjusted accordingly	a Wilcoxon Rank test with $p < 0.05$ was conducted to evaluate the enhancement of players' experiences in the game with DDA compared to the version without DDA
[64]	2018	C	Creating a believable intelligent agent to adjust the difficulty	Conducting a Friedman test on the suggested agent's believability ($p < 0.05$) to see if it could dynamically modify the difficulty while restraining its unnatural actions
[65]	2018	J	Two difficulty adjustment approaches are compared: heuristic-triggered DDA, which changed difficulty levels based on static rules and player-kill ratio, and EEG-triggered DDA, which changed difficulty levels depending on players' excitement levels	Using the EEG-triggered DDA approach, an ANOVA test with $F = 5.25$ revealed that gamers had better experiences with higher degrees of excitement
[66]	2017	J	Proposing a player model based on an expert's strategy, performance, and eye gaze and converting low-level gaming data to three components of flow theory, namely experience, exploration, and attention, and adjusting the game's difficulty based on computed values for the experience, exploration, and attention	The results demonstrated that the proposed DDA approach could maintain a high degree of player satisfaction (multiple statistical analyses have been conducted)

Table 6 (continued)

Title	Year	Publication	Target	Evaluation and result
[67]	2017	C	Using Rhythm-based procedural content generation and defining the difficulty of generated levels based on a mix of player performance and attention data gathered from EEG signals	Adapting the game's difficulty to players with varying levels of experience step by step (no statistical analysis reported)
[68]	2017	C	Changing the spawn rates of enemies based on the number of players' health points to adjust the game's difficulty	The proposed adjustment approach has increased the enjoyment experienced by participants by 40% (no statistical analysis reported)
[69]	2017	C	Comparing the impact of three different ways of difficulty adjustment on a player's experience: 1- no DDA: a control condition in which there is no DDA implementation, 2- rDDA: a system-oriented ramping DDA in which the difficulty automatically increases based on the player's performance, and 3- pDDA: a player-oriented DDA in which players had direct control over the difficulty level	With $F=9.33$, the MANOVA test was used to analyze the effect of the three adjustment techniques (no-DDA, rDDA, and pDDA) on the variables of the Flow State Scale (FSS) questionnaire. Compared to no-DDA systems, DDA systems improved players' experiences; however, player-oriented DDA gives players a stronger sense of control than ramping DDA
[70]	2017	C	Changing game difficulty to maximize the number of rounds played as a metric of player engagement	When compared to the default version, employing the with-DDA version improves player engagement ($p < 0.05$)
[7]	2017	C	Designing multiple artificial intelligent agents with different action selection policies controlled the opponent's difficulty level	Each agent's win rate is reported
[71]	2017	J	Introducing an evaluation function and calculating the difference between the player and the opponent's values. If the computed difference exceeds a certain threshold, the game's difficulty changes	Providing an acceptable level of immersion
[72]	2016	J	Using two bands of the EEG signal (frontal theta and parietal alpha) as indicators for evaluating motivation and effort, then comparing the effort and motivation to a baseline value and modifying difficulty based on the difference	Players' experiences were compared in three types of bio cybernetics adjustment loops: conservative, liberal, and moderate. The conservative system has resulted in the most significant demand
[73]	2016	J	Learning an optimal policy to make the game more difficult for players and achieving the optimal challenge level for multiple players in real-time using Distributed Constraint Optimization (DCO)	Finding an adequate difficulty level for all players in a shared simulation space
[14]	2016	J	Selecting the best difficulty adjustment method based on players' personality traits that maximized engagement and gameplay duration	ANOVA test results showed that selecting a difficulty adjustment method based on players' personality traits increases players' enjoyment with $F=174.4$ and gameplay duration with $F=31.5$ compared to a technique that does not consider personality traits

Table 6 (continued)

Title	Year	Publication	Target	Evaluation and result
[74]	2016	C	A comparison of two approaches for adjusting difficulty: 1- a time-based strategy that gradually increases the difficulty over 5 min. 2- a mental state-based method that gradually increases the difficulty depending on the amount of attention	The ability of the mental state-based strategy to change the level of difficulty was revealed (no statistical analysis reported)
[16]	2016	C	Assigning the appropriate level of difficulty to each player using a genetic algorithm	Results reported that the player's enjoyment increased compared to randomly adjusted levels across the adaptive levels
[75]	2016	C	Comparing the effect of a player-oriented adjustment method and an automated DDA method on players' experience	Players preferred the player-oriented method, but they were pleased to have DDA included
[76]	2016	C	Levels are generated using a genetic algorithm, and the difficulty of levels is adjusted using a player performance-based fitness function	Reporting the correlation between difficulty impact factor and win percentage
[21]	2015	J	Adapting game difficulty by keeping players' arousal at a constant level using a classic PID controller	A lower error has been reported when using a PID controller (MSE= 2.61) compared to a proportional (P) controller (MSE=2.75)
[77]	2015	J	Changing the game environment to reduce the likelihood of quitting the game, which is calculated based on the state sequence	Conducting a t-test with $t=2.30$ to compare the effect of using adaptive and non-adaptive versions of a board game on player's engagement through the Games Engagement Questionnaire (GEQ)
[78]	2015	J	Changing the game's difficulty level by relocating, tinting, or altering game elements to make them more or less noticeable, based on the saliency map and attention prediction	Reporting on the relationship between the game's level completion time and the topology of the game elements and the viability of utilizing this model to change the level of difficulty
[79]	2015	J	A player-oriented difficulty adjustment and an automatic adjustment approach that raises or decreases N based on the player's error were compared in an N-back memory game	In the player-oriented version, there were greater measures for session frequency ($F=6.4$), duration ($F=5.01$), and motivation ($F=12.67$) than in the automatic version
[17]	2015	J	Proposing an adjustment method called DDA-visual in which changed visual elements such as background, foreground, and animation based on the success or failure of rounds	Compared to the non-visual DDA, the DDA-Visual version reported a higher number of rounds played ($p<0.05$) and more enjoyment ($p<0.05$)
[80]	2015	J	Constrained optimization was used to estimate temporal player performance using tensor factorization and modifying contents to minimize the desired and predicted performance difference	Results showed that tensor factorization could be used for game adaptation and accurately predict player learning trends

Table 6 (continued)

Title	Year	Publication	Target	Evaluation and result
[81]	2015	C	Changing game difficulty by modifying the game timer	Using the Immersive Experience Questionnaire (IEQ), participants reported increased immersion ($p < 0.05$) after playing the game with an altered timer compared to the standard timer
[82]	2015	C	Predicting the player's skill and changing eight distinct levels according to the player's skill	The post-test showed improved performance over the pre-test, and the game can help keep players motivated
[83]	2015	C	Asking players' opinions about five different scenarios for changing the game's difficulty and using this information and the player's estimated profile for level adjustment	The adaptive game provided a higher level of satisfaction than the non-adaptive game
[84]	2014	J	The game's difficulty is divided into N levels, with N being adjusted based on the player's performance	Enhancing game experience using the DDA mechanism compared to a static difficulty adjustment version
[85]	2014	C	Changing the difficulty of an exercise to get a balanced level with therapy protocol based on players' success rates	Using the Snellen Vision Test, the adaptive system indicated a 2–3 level improvement in visual acuity
[86]	2014	C	Comparing the effect of using 3 versions of the game: 1- without-DDA, 2- with DDA but participants are unaware of its presence, 3- with DDA and participants are aware	The DDA version improves the player's performance and reduces the challenge; player awareness of DDA's presence reduces its impact
[87]	2014	C	Investigating the effect of four difficulty adjustment techniques on player's experience	Improving performance and reporting positive affect
[88]	2014	C	Comparing the effect of three kinds of difficulty adaptation on player's performance and enjoyment: 1- a user-guided adaptation based on self-determination theory, 2- an automated task-guided adaptation based on flow theory, and 3- random adaptation	In the user-guided version, there was more enjoyment ($p < 0.05$), while in the task-guided version, there was more performance ($p < 0.05$)
[89]	2013	J	Investigating the difference between experienced or casual players' preferences in selecting dynamic or static difficulty adjustment methods	According to the findings, it would be preferable to adjust the difficulty based on the player's experience rather than their ability, as this would increase enjoyment
[90]	2013	J	Using HMM to recognize facial expressions and the Kalman Filter to predict players' experiences, and then modifying game difficulty based on the player's experience	Using DDA based on facial expression recognition improved the player's experience compared to DDA based on the player's performance (no statistical analysis reported)

Table 6 (continued)

Title	Year	Publication	Target	Evaluation and result
[91]	2013	J	Using gameplay data and head movement features to train a neural network to predict player engagements and then designing levels that maximize the network's output	Results showed that players' visual reaction features can be used to predict challenges with an accuracy of 88.88%, annoyance at 92.5%, and engagement at 91.27%
[22]	2012	J	Using the player's accuracy and response time to determine if the player is risky or cautious, and adapting the game based on the risk profile	Results showed that Particle Filter provides an efficient approach for adapting game parameters
[92]	2012	J	Classifying players' behavior patterns to adjust the game difficulty	Classifier accuracy can be improved by compressing data without reduction and tackling the sparseness problem of gameplay data with Higher-Order Singular Value Decomposition (HOSVD)
[3]	2012	C	Predicting temporal player's performance using a data-driven approach and minimizing the difference between predicted performance and desired performance curve to apply difficulty adjustment	Results showed that the temporal model is capable of successfully capturing data fluctuation
[93]	2011	C	Modeling players' types using the Myers-Briggs Type Indicator (MBTI) questionnaire and adapting game contents accordingly	When the game adapts to the personality features of the players, the results show a higher level of immersion ($p=0.08$, $p<0.1$)
[94]	2010	C	Players' profiles are determined by asking them to complete a questionnaire and then adjusting the difficulty based on their performance and profile	Improving player's experience has been reported
[95]	2009	J	Predicting player's anxiety level using a regression tree trained on physiological data and then adjusting the specified level of difficulty using a state flow diagram	When comparing the affect-based adjustment approach to the performance-based adjustment method, the affect-based adjustment method improved the player's performance and satisfaction ($p<0.05$)
[15]	2009	C	An intelligent opponent was created that can adjust an offensive factor by observing and assessing players' movements to make the game more difficult for players	The results showed that while playing against an intelligent opponent who can scale the level of difficulty, the player's satisfaction increases (no statistical analysis reported)
[96]	2009	C	Changing game difficulty based on players' stress level, which was assessed using facial physiology	When compared to a fixed difficulty adjustment version, the proposed difficulty adjustment approach maintains player interest while playing a game

Appendix 2

Table 7 Lists of included articles that focused just on building a player model

Title	Year	Publication	Target	Evaluation and result
[102]	2021	J	Using the concept of transfer learning to solve the "cold start" problem and predict players' affective states in terms of valence, arousal, dominance, and liking	Reporting F-measure for LSTM-DNN (0.69) compared to Support Vector Machine (SVM) (0.42) and Decision tree (0.37) on DEAP dataset
[105]	2020	C	Using Deep Player Behavior Modeling (DPBM) to substitute disconnected players in online matches by learning bots that emulate the individual behavior of a particular player to reduce player's drop-off due to lost network connections in the online game	According to the results of the in-game bot detection survey, 85.48% of participants could not differentiate between human and DPBM behavior
[106]	2020	J	The game attributes were clustered using the K-mean approach, with each cluster representing one of the four-player categories. The cluster name is then used as a label for decision tree training and player classification	The accuracy of the proposed classification's compatibility with player profiles was reported to be between 75 and 80%
[23]	2019	C	Proposing a deep learning-based player behavior model based on massively multiplayer online game datasets	The results indicated that deep learning could produce more accurate models of individual player behavior with 73.4% accuracy than HMM with 48% accuracy and decision trees with 61.7% accuracy
[107]	2018	C	Proposing a model for predicting players' abilities based on less than a minute's worth of data from keyboard and mouse inputs	The trained model's accuracy in predicting players' skills was reported to be 77.1%
[108]	2017	J	Estimating the player's style and expertise based on the sequence of actions performed in the game	The HMM model was shown to be 70.13% accurate in predicting game expertise and different play styles
[109]	2017	C	Based on the segmentation of gaming traces, sequential machine-learning is used to predict players' play styles	Sequential machine-learning algorithms were found to be more accurate than non-sequential techniques
[110]	2015	C	Clustering players' styles based on reducing the dimensionality of gameplay log data	Linear Discriminant Analysis (LDA) is the most effective method of determining the difference between players' styles
[111]	2012	J	Predicting a player's skill level and performance	The HMM model was found to be more accurate than the LSTM and Kalman Filter models

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